II. Entanglement

1. Introduction

Consider a bipartite state $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$.

We say that $|\psi\rangle$ is a product state if it can be written as

$$|\psi\rangle = |\phi^A\rangle \otimes |\phi^B\rangle$$

for some $|\phi^A\rangle \in \mathcal{H}_A$, $|\phi^B\rangle \in \mathcal{H}_B$.

If $|\psi\rangle$ is not a product state, we say that $|\psi\rangle$ is entangled.

How can we tell if a state is a product state or entangled?

Consider Schmidt decomposition of $|\psi\rangle$:

$$|\psi\rangle = \sum_{i=1}^{r} \lambda_i |\phi_i^A\rangle \otimes |\phi_i^B\rangle$$

(orthonormal)

with r the Schmidt rank.

- $r = 1 \Rightarrow |\psi\rangle = |\phi_1^A\rangle \otimes |\phi_1^B\rangle \Rightarrow |\psi\rangle$ product.
- $|\psi\rangle$ product $\Rightarrow |\psi\rangle = |\phi^A\rangle \otimes |\phi^B\rangle$ is Schmidt decomposition $\Rightarrow r = 1$.

I.e.: Product states have Schmidt rank $r = 1$.
Entangled states have Schmidt rank $r \geq 2$.

In particular, states such as

$$|\psi^-\rangle = \frac{1}{\sqrt{2}} (|0\rangle|1\rangle - |1\rangle|0\rangle)$$

or

$$|\Omega\rangle = \frac{1}{\sqrt{d}} \sum_{i=1}^{d} |i, i\rangle$$

are entangled.

Product states: Can describe any action of A & B independently, and state

$$|\psi\rangle = |\phi^A\rangle \otimes |\phi^B\rangle \longmapsto (\Pi_A \otimes N_B)|\psi\rangle$$
$$= \Pi_A |\phi^A\rangle \otimes N_B |\phi^B\rangle$$

stays a product state.

Entangled states:

Generally cannot describe actions of A & B independently – e.g., meas. outcomes in Schmidt basis will be perfectly correlated.

Can we use these correlations for non-trivial tasks?

Questions in entanglement theory:

- How non-classical are entangled states?
- What can we do with entangled states?
  (i.e., are they a resource?)
- How can we quantify the amount of entanglement (e.g. in terms of usefulness)?
- Are there different types of entangled states?
- How can we manipulate entangled states?
- What about mixed state entanglement?

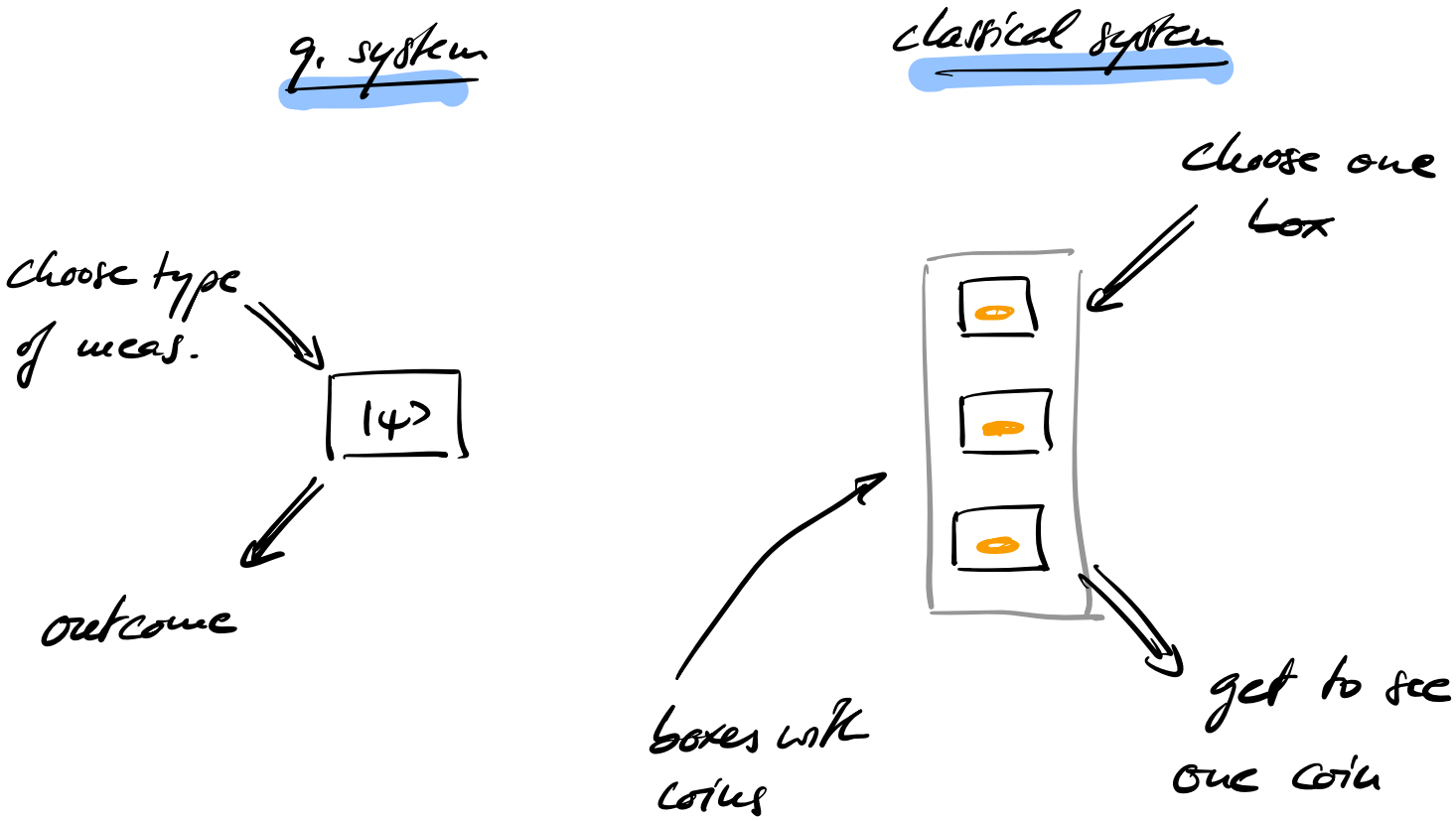
2. Bell inequalities

"How non-classical are entangled states?"

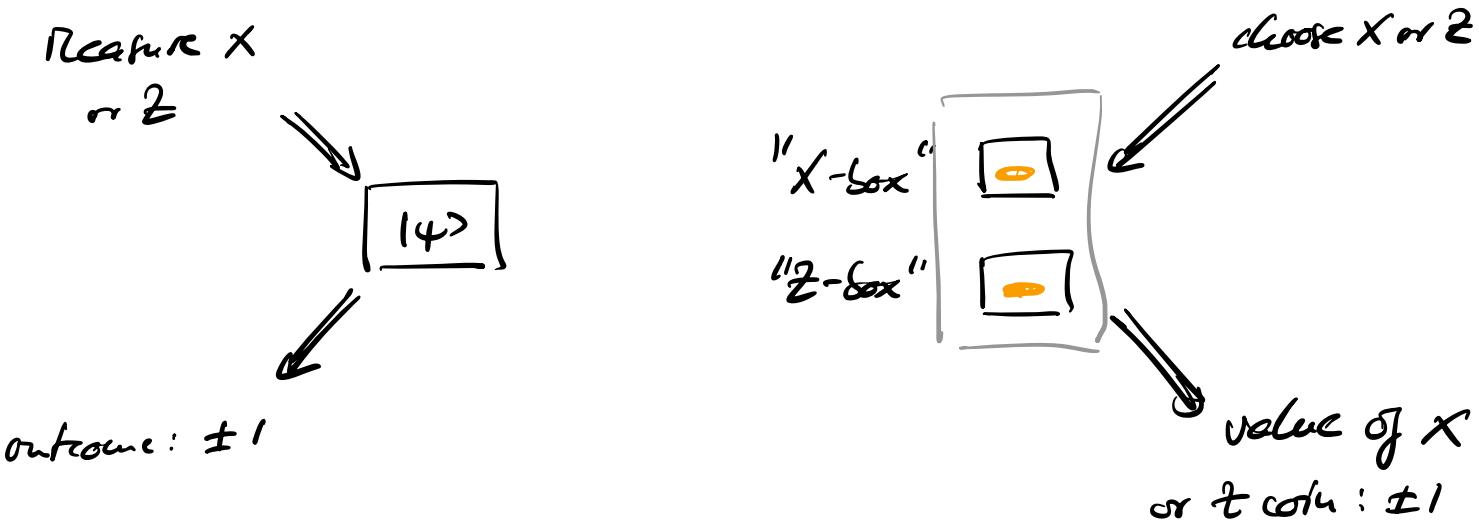
— i.e., how different from objects which follow probability theory, such as coins.

a) Setting

Compare quantum theory vs. coins (= prob. theory) in elementary setting:



E.g.:



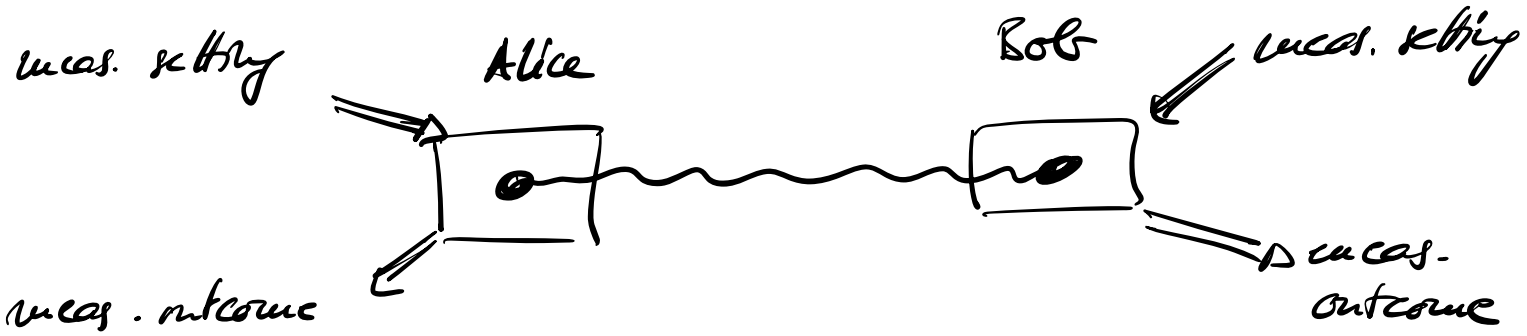
Do the predictions in these two models differ?

For single quantum system: NO!

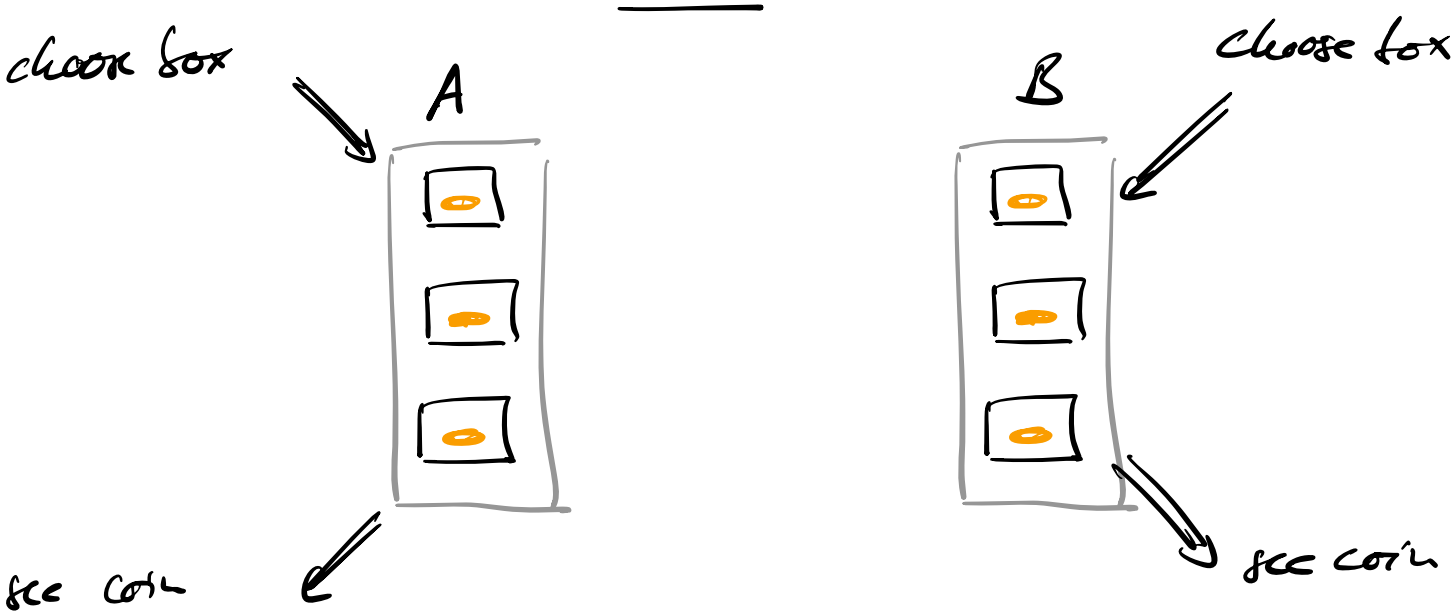
Boxes can (1) store probabilities for any meas. &
(2) update themselves – if needed – following
the laws of q. theory.
(NB: Can even be done storing finite amount
of information.)

What about bipartite systems?

Compare quantum:



vs. classical

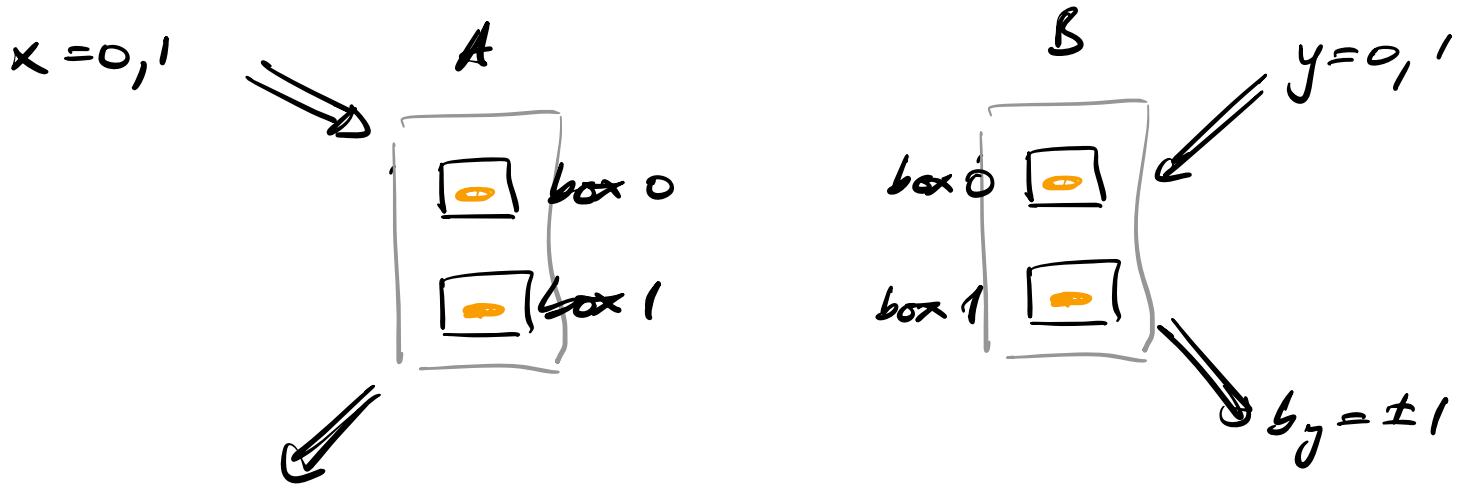


Boxes prepared in advance (by "referee") according to some fixed rule (which can involve randomness), but then, the boxes are separated, i.e., the two of A & B cannot exchange information ("locality").

Are there predictions for quantum systems which cannot be reproduced with classical coins?

b) The CHSH inequality

x: which box



a_x: outcome +1 (head)
or −1 (tail)

$a_0 = \pm 1$: value of coin 0 in box 0

$a_1 = \pm 1$: — " — box 1

$b_0 = \pm 1$: val. of coin in box 0

$b_1 = \pm 1$: val of coin in box 1

In each round: Coins prepared by referee R
following fixed procedure (including randomness),
then A & B "measure" a coin.

In any round, since a_0, a_1, b_0, b_1 have well-defined values, and $a_0 = \pm 1$, $a_1 = \pm 1$, $b_0 = \pm 1$, $b_1 = \pm 1$

$$C = (a_0 + a_1)\, b_0 + (a_0 - a_1)\, b_1 = \pm 2$$

one of these must be 0,
the other is ± 2.

If we average C over many rounds, we thus get for the average value $\langle C \rangle$:

$$|\langle C \rangle| \leq \langle |C| \rangle = 2$$

and, of course,

$$C = a_0 b_0 + a_1 b_0 + a_0 b_1 - a_1 b_1 .$$

We thus find that

$$\boxed{|\langle a_0 b_0 \rangle + \langle a_1 b_0 \rangle + \langle a_0 b_1 \rangle - \langle a_1 b_1 \rangle| \leq 2}$$

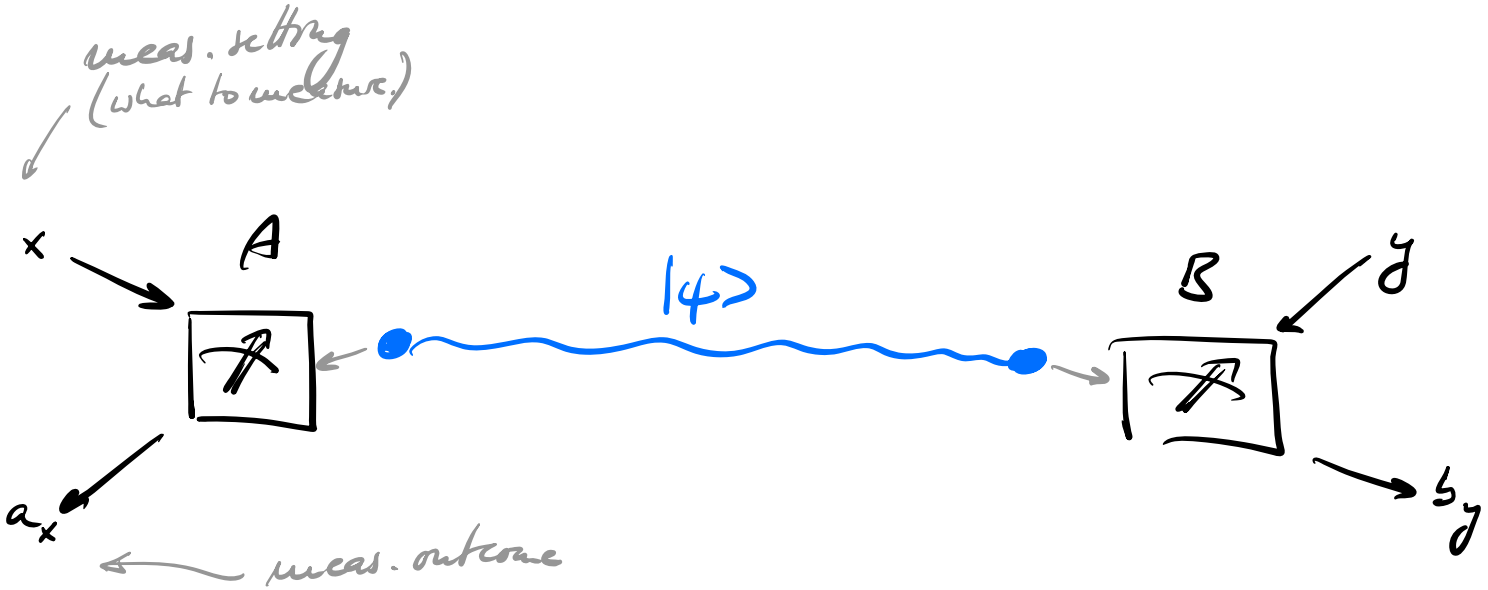
"CHSH inequality" (Clauser, Horne, Shimony, Holt)

$\Rightarrow$ Must be satisfied by any local classical theory.

Note: Even though A & B can each only measure one of a_0/a_1 or b_0/b_1 per round, they can estimate $\langle a_x b_y \rangle$ $\forall x,y$ by measuring many rounds with all possible settings $x=0,1$; $y=0,1$.

c) Quantum violation of CHSH inequality

Quantum version



- referee R distributes bipartite state $|\psi\rangle$.
- A & B perform a measurement dep. on x/y, w/ outcome a_x/b_y.

Setup:

- $|\psi\rangle = |\psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$
- A & B do projective measurement along axes $\vec{u}_x$ and $\vec{u}_y$, i.e. they measure the operators $\vec{u}_x \cdot \vec{\sigma}^A$ and $\vec{u}_y \cdot \vec{\sigma}^B$.
- We will specify $\vec{u}_x$ and $\vec{u}_y$ later on.

It holds that $\left(\vec{\sigma}^A + \vec{\sigma}^B\right)|\psi^-\rangle = 0$

$\vec{\sigma}^A \equiv \vec{\sigma}_A \otimes \mathbb{1}_B$ $\qquad$ $\vec{\sigma}^B \equiv \mathbb{1}_A \otimes \vec{\sigma}^B$

(i.e. $\left(\sigma_\alpha^A + \sigma_\alpha^B\right)|\psi^-\rangle = 0 \quad \forall \alpha = x,y,z$;

by direct inspection, or since $|\psi^-\rangle$ has spin 0).

$$\Rightarrow \langle\psi^-| \left(\vec{\sigma}^A \cdot \vec{u}\right)\left(\vec{\sigma}^B \cdot \vec{w}\right)|\psi^-\rangle =$$

$\left(\vec{\sigma}^B \cdot \vec{w}\right)|\psi^-\rangle = -\vec{\sigma}^A \cdot \vec{w}\,|\psi^-\rangle$ from the above

$$= -\langle\psi^-| \left(\vec{\sigma}^A \cdot \vec{u}\right)\left(\vec{\sigma}^A \cdot \vec{w}\right)|\psi^-\rangle$$

$$= -\sum_{kl} u_k w_l \langle\psi^-|\sigma_k^A \sigma_l^A|\psi^-\rangle$$

$\langle\psi^-|\sigma_k^A \sigma_l^A|\psi^-\rangle = \mathrm{tr}\left[\rho_A \sigma_k^A \sigma_l^A\right]$

$\rho_A = \frac{1}{2}\mathbb{1}$, e.g. from Schmidt dec.

$= \frac{1}{2}\mathrm{tr}\left[\sigma_k^A \sigma_l^A\right] = \delta_{kl}$

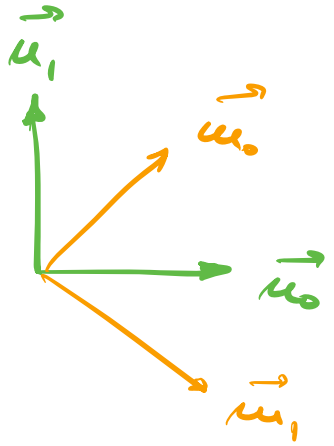
$$= -\sum_k u_k w_k = -\vec{u}\cdot\vec{w} = -\cos\theta$$

angle between $\vec{u}$ and $\vec{w}$.

Violation of CHSH inequality in quantum theory!

Take $a_x \longleftrightarrow \vec{u}_x \cdot \vec{\sigma}^A$

$b_y \longleftrightarrow \vec{w}_y \cdot \vec{\sigma}^B$



XZ plane

It holds that

± 1 ± 1 eigenvalues ± 1 eigenvalues ± 1

$$\langle a_x b_y \rangle = \langle \psi^- | (\vec{u}_x \cdot \vec{\sigma}^A) \otimes (\vec{w}_y \cdot \vec{\sigma}^B) | \psi^- \rangle$$

$$= -\cos\theta$$

$$\Rightarrow \langle a_0 b_0 \rangle = \langle a_0 b_1 \rangle = \langle a_1 b_0 \rangle = -\frac{1}{\sqrt{2}}$$

$$\langle a_1 b_1 \rangle = +\frac{1}{\sqrt{2}}$$

$$\Rightarrow \left| \langle a_0 b_0 \rangle + \langle a_1 b_0 \rangle + \langle a_0 b_1 \rangle - \langle a_1 b_1 \rangle \right| = 2\sqrt{2} > 2 \; !$$

CHSH inequality violated!

Quantum theory can exhibit correlations which classical local theories cannot!

Formally, what were the assumptions of our classical theory?

① Realism: Outcomes of measurements are "elements of reality" – i.e., they have pre-determined values even prior to measurement.

② Locality: A & B's boxes cannot communicate once distributed.

⇒ Quantum mechanical predictions are incompatible with any local and realistic theory – we need to give up either Locality or realism.

Such inequalities, which must be satisfied by a classical local model, but can be (potentially) violated by quantum theory, are called Bell

inequalities. They can be used to certify that a system behaves quantum mechanically (i.e. non-classically): If we measure a value of the Bell inequality, we know that the system cannot be described classically and must thus be quantum mechanical.

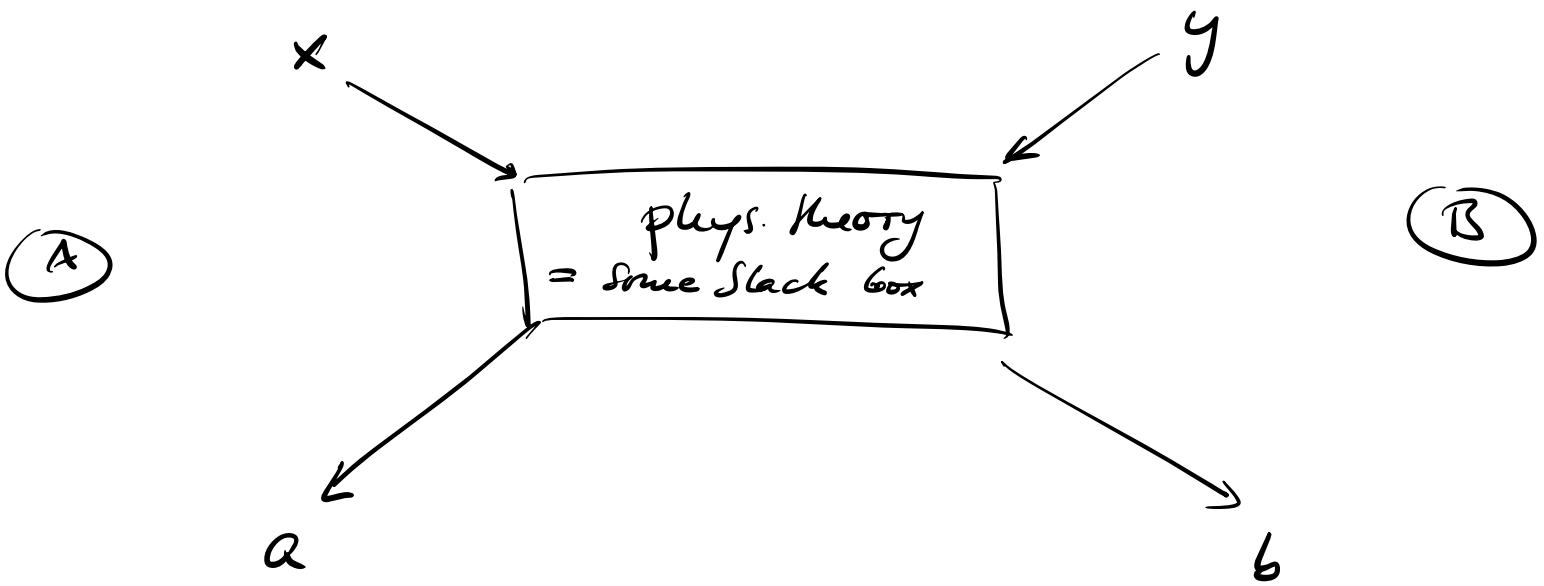
Note: The value $\langle C \rangle = 2\sqrt{2}$ is optimal (i.e., maximal) within quantum theory.

But: With a general local theory, $|\langle C \rangle| = 4$ can be obtained: QM is more restrictive than general local theories.

(→ Homework)

d) Formal setup and local hidden variable theories

Formal setup for physical theories in bipartite setting:



A: input x (meas. setting),
output a (meas. result)

B: input y, output b.

Any physical theory is characterized by a conditional probability distribution

$$P(a,b|x,y)$$

to obtain a & b given x & y, where

$$\sum_{a,b} P(a,b|x,y) = 1 \quad \forall x,y.$$

Question: Which $P(a,b|x,y)$ are consistent with a given physical theory?

Classical physics:

"local hidden-variable (LHV) model":

All outcomes are pre-determined by some "hidden" variable λ, which is chosen according to some distribution & give to A & B, who act independently (i.e., locally) conditioned on λ.

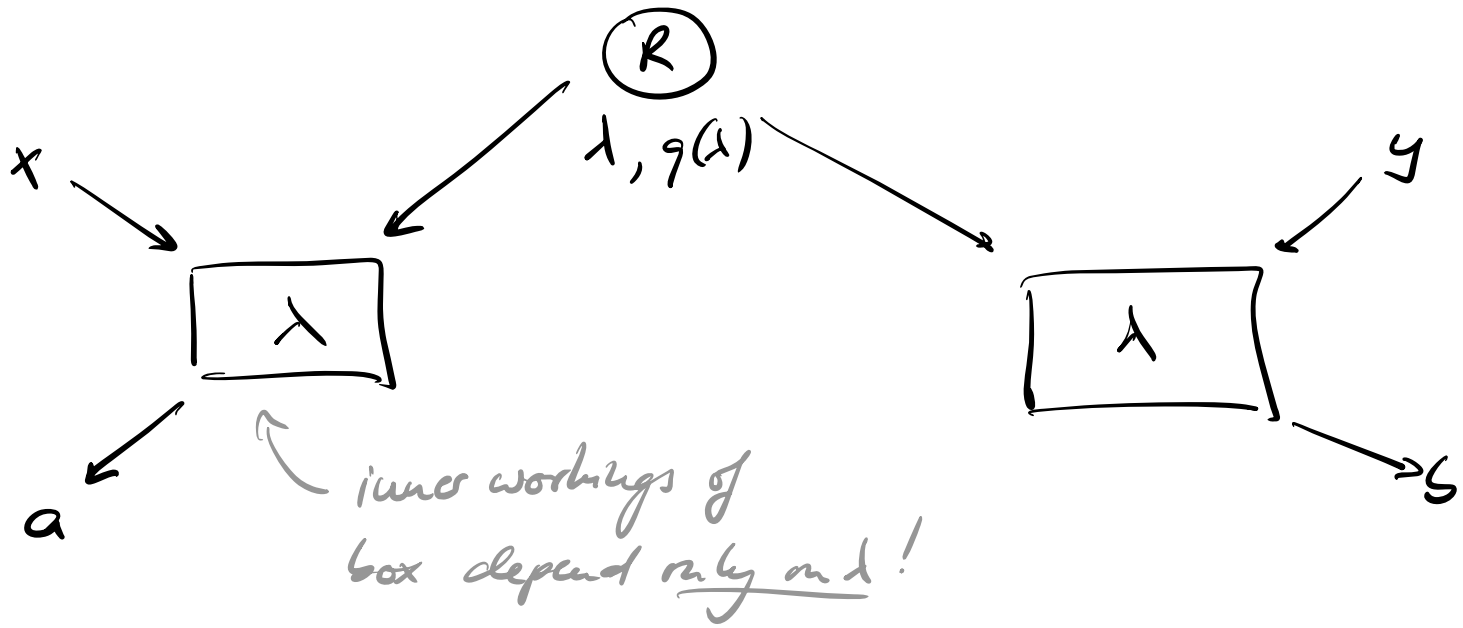
("local realism": Outcomes exist indep. of meas. -realism- and no (faster-than-light) communication between A & B — locality.)

i.e.:

$$\circledast \quad P(a,b|x,y) = \sum_{\lambda} q(\lambda)\, P^{A}_{\lambda}(a|x)\, P^{B}_{\lambda}(b|y)$$

prob. distr. over λ

can be made deterministic by putting all randomness in λ & $q(\lambda)$.



How have we been using the LHV form ⊛ above in the derivation of the CHSH inequality?

$$\langle C \rangle = \langle a_0 b_0 \rangle + \langle a_1 b_0 \rangle + \langle a_0 b_1 \rangle - \langle a_1 b_1 \rangle$$

$$= \sum_{x,y} (-1)^{xy} \langle a_x b_y \rangle$$

$$= \sum_{x,y} (-1)^{xy} \left[\sum_{a_x, b_y} a_x b_y \; P(a_x, b_y | x, y) \right]$$

$$= \sum_{\lambda} q(\lambda) P_\lambda^A(a_x|x) P_\lambda^B(b_y|y)$$

$$= \sum_{\lambda} q(\lambda) \underbrace{\sum_{x,y} (-1)^{xy} \left[\sum_{a_x, b_y} a_x b_y \; P_\lambda^A(a_x|x) P_\lambda^B(b_y|y) \right]}_{=: \epsilon_\lambda}$$

Then, $|\langle C \rangle| \leq \sum_{\lambda} q(\lambda) \, |\epsilon_\lambda|$, and

$$|\epsilon_\lambda| = \left| \sum_{x,y} (-1)^{xy} \underbrace{\left(\sum_{a_x} a_x \, P_\lambda^A(a_x|x) \right)}_{= \langle a_x \rangle_\lambda} \underbrace{\left(\sum_{b_y} b_y \, P_\lambda^B(b_y|y) \right)}_{= \langle b_y \rangle_\lambda} \right|$$

$$= \left| \left(\langle a_0 \rangle_\lambda + \langle a_1 \rangle_\lambda \right) \langle b_0 \rangle_\lambda + \left(\langle a_0 \rangle_\lambda - \langle a_1 \rangle_\lambda \right) \langle b_1 \rangle_\lambda \right|$$

$$\leq \left| \langle a_0 \rangle_\lambda + \langle a_1 \rangle_\lambda \right| \underbrace{\left| \langle b_0 \rangle_\lambda \right|}_{\leq 1} + \left| \langle a_0 \rangle_\lambda - \langle a_1 \rangle_\lambda \right| \underbrace{\left| \langle b_1 \rangle_\lambda \right|}_{\leq 1}$$

$$\leq \left| \langle a_0 \rangle_\lambda + \langle a_1 \rangle_\lambda \right| + \left| \langle a_0 \rangle_\lambda - \langle a_1 \rangle_\lambda \right|$$

$$\leq 2 \max \left\{ \left| \langle a_0 \rangle_\lambda \right|, \left| \langle a_1 \rangle_\lambda \right| \right\} \leq 2.$$

Where have we used the LHV condition ⊛?

→ in factorizing

$$\langle a_x b_y \rangle_\lambda = \sum_{a_x, b_y} a_x b_y \; P(a_x, b_y | x, y)$$

$$= \left(\sum_{a_x} a_x \; P^A_\lambda(a_x | x) \right) \left(\sum_{b_y} b_y \; P^B_\lambda(b_y | y) \right)$$

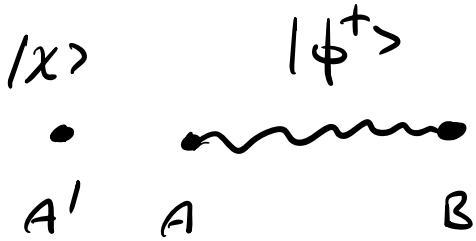
$$= \langle a_x \rangle_\lambda \langle b_y \rangle_\lambda$$

— otherwise, it does not make sense to talk about $\langle a_x \rangle$ indep. of the value of y, and $\langle a_0 b_0 + a_0 b_1 + a_1 b_0 - a_1 b_1 \rangle$ cannot be factorized (either as expectation values $\langle a_i \rangle \langle b_j \rangle$, or classical variables — cons — sketch is in essence the same.)

3. Applications of entanglement:
Teleportation and dense coding

a) Teleportation

Setup:



- A & B share entangled state $|\phi^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB})$
- A has unknown quantum state

$$|\chi\rangle_{A'} = a|0\rangle_{A'} + b|1\rangle_{A'}$$

(Could e.g. also be part of a larger system → Ignorance!)

- A & B cannot (reliably*) transmit quantum states, but can communicate classically "for free".

* If the line is unreliable, A&B can still use it to create entangled states $|\phi^+\rangle$, e.g. by repeat-until-success, or entanglement distillation (→ later!), or using "quantum repeaters" (→ later!)

Question: Can A get $|\chi\rangle$ (safely) to B?

Problem: Any measurement of $|X\rangle$ would only reveal partial information, yet destroy state!

Solution: Quantum Teleportation!

Teleportation Protocol:

1. A performs measurement on $A'A$ in Bell basis

$$|\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$|\phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) = (Z \otimes I)|\phi^+\rangle = (I \otimes Z)|\phi^+\rangle$$

$$|\psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) = (X \otimes I)|\phi^+\rangle = (I \otimes X)|\phi^+\rangle$$

$$|\psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) = (iX \otimes I)|\phi^+\rangle = (I \otimes XZ)|\phi^+\rangle$$

We also write the four Bell states as

$$|\phi_{\alpha\beta}\rangle = (Z^\alpha X^\beta \otimes I)|\phi^+\rangle = (I \otimes X^\beta Z^\alpha)|\phi^+\rangle$$

$\alpha, \beta = 0, 1$

Outcome probabilities for meas. outcome $|\phi_{\alpha\beta}\rangle$:

$$\rho_A = \mathrm{tr}_B\left[|\phi^+\rangle\langle\phi^+|_{AB}\right] = \frac{1}{2} I_A \quad \leftarrow \text{state of } A.$$

$$p_{\alpha\beta} = \langle \phi_{\alpha\beta}| \, |\chi\rangle\langle\chi|_{A'} \otimes \tfrac{1}{2} I_A \, |\phi_{\alpha\beta}\rangle$$

$$= \frac{1}{2}\,\mathrm{tr}\left[\left(|\chi\rangle\langle\chi|_{A'} \otimes I_A\right) |\phi_{\alpha\beta}\rangle\langle\phi_{\alpha\beta}|\right]$$

$$= \frac{1}{2}\,\mathrm{tr}_{A'}\left[\, |\chi\rangle\langle\chi|_{A'} \cdot \underbrace{\mathrm{tr}_A\left[|\phi_{\alpha\beta}\rangle\langle\phi_{\alpha\beta}|\right]}_{= \frac{1}{2} I_{A'}}\right]$$

$$= \frac{1}{2}\,\mathrm{tr}\left[|\chi\rangle\langle\chi|_{A'} \cdot \frac{1}{2} I_{A'}\right]$$

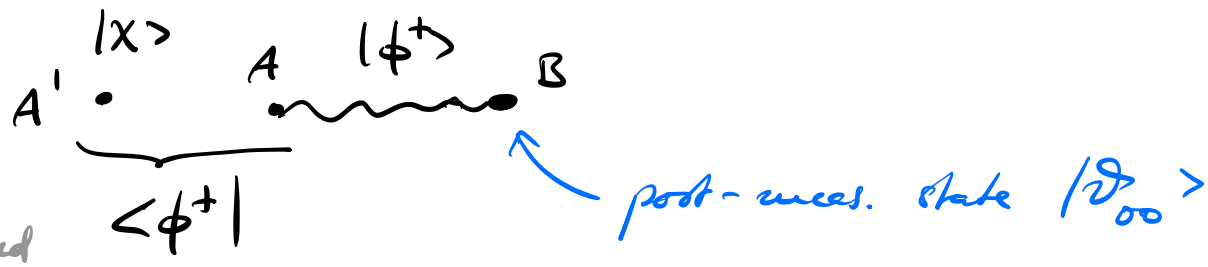
$$= \frac{1}{4}$$

$\Rightarrow$ equal probability $p_{\alpha\beta} = 1/4$ for all outcomes.

(This is good – if $p_{\alpha\beta}$ would depend on $|\chi\rangle$, it would reveal information on $|\chi\rangle$ and thus perturb the state!)

What is the state of B after the measurement?

i) Outcome $|\phi^+\rangle = |\phi_{00}\rangle$:



unnormalized

$$|\tilde{\vartheta}_{00}\rangle = \langle\phi^+|_{A'A}\left(|\chi\rangle_{A'} \otimes |\phi^+\rangle_{AB}\right)$$

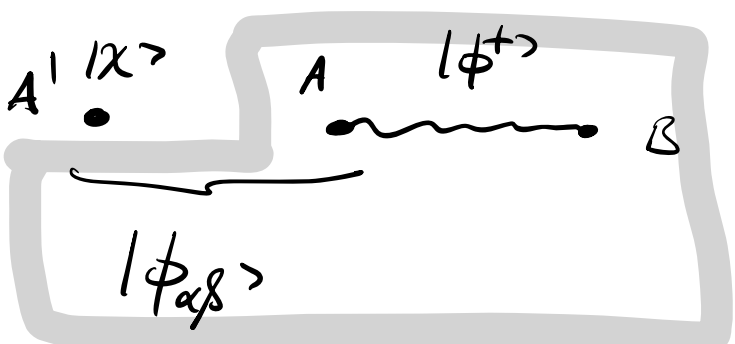
$$= \frac{1}{2}\left(\langle 00|_{A'A} + \langle 11|_{A'A}\right)\left(\underbrace{\left(a|0\rangle_{A'} + b|1\rangle_{A'}\right)}_{= a\langle 0|_A + b\langle 1|_A} \otimes \left(|00\rangle_{AB} + |11\rangle_{AB}\right)\right)$$

$$= \frac{1}{2}\left(a|0\rangle_B + b|1\rangle_B\right) \qquad (*)$$

$\Rightarrow$ State $|\vartheta_{00}\rangle = |\chi\rangle$ appears at B!

(works with 25% probability.)

ii) What about the other outcomes?



First consider $\langle \phi_{\alpha\beta}|_{A'A} |\phi^+\rangle_{AB}$ — worked

grey above:

$$\langle \phi_{\alpha\beta}|_{A'A} |\phi^+\rangle_{AB} = \langle \phi^+|_{A'A} \left(I_{A'} \otimes z_A^\alpha x_A^\beta\right) |\phi^+\rangle_{AB}$$

$$= \langle \phi^+|_{A'A} \left(z_A^\alpha x_A^\beta \otimes I_B\right) |\phi^+\rangle_{AB}$$

$$= \langle \phi^+|_{A'A} \left(I_A \otimes x_B^\beta z_B^\alpha\right) |\phi^+\rangle_{AB}$$

$$= x_B^\beta z_B^\alpha \langle \phi^+|_{A'A} |\phi^+\rangle_{AB}$$

Now combine with observation ⊛ in part i)

$$|\tilde{\vartheta}_{\alpha\beta}\rangle = \langle \phi_{\alpha\beta}|_{A'A} \left(|\chi\rangle_{A'} \otimes |\phi^+\rangle_{AB}\right)$$

$$= x_B^\beta z_B^\alpha \langle \phi^+|_{A'A} \left(|\chi\rangle_{A'} \otimes |\phi^+\rangle_{AB}\right)$$

$$\overset{⊛}{=} \tfrac{1}{2}|\chi\rangle_B$$

$$= \tfrac{1}{2} x^\beta z^\alpha |\chi\rangle_B .$$

$\Rightarrow$ After A's measurement, B obtains $|\vartheta_{\alpha\beta}\rangle = X^\beta Z^\alpha |\chi\rangle$

with probability 1/4 each.

$\Rightarrow$ average state of B — without learning meas. result — is $\frac{1}{4}\sum X^\beta Z^\alpha |\chi\chi|\chi Z^\alpha X^\beta = \frac{1}{2} I$.

i.e.: Bob has no information about $|\chi\rangle$

(in fact: same state as without meas.)

(2) A communicates meas. outcome (α,β) to B, and

(3) B applies $(X^\beta Z^\alpha)^\dagger$ to their state

$\Rightarrow$ Bob obtains

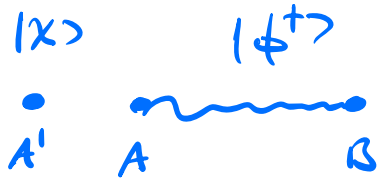
$$(X^\beta Z^\alpha)^\dagger |\vartheta_{\alpha\beta}\rangle = (X^\beta Z^\alpha)^\dagger (X^\beta Z^\alpha)|\chi\rangle = \underline{|\chi\rangle}$$

$\Rightarrow$ Bob obtains $|\chi\rangle$ with probability 1!

$\Rightarrow$ State $|\chi\rangle$ has been teleported to B.

Notes:
- No faster-than-light communication (any. state of B is $\frac{1}{2} I$ prior to receiving (α,β) — which has finite transm. speed.)

- Communicating 1 qubit requires 1 "e-bit"
  (= a max. entangled state $|\phi^+\rangle$ of 1+1 qubit)
  + 2 bits of classical communication ("c-bits")



Teleportation protocol – summary:

1. Prepare A, A' in $|\phi_{\alpha\beta}\rangle$ basis.
2. Communicate $(\alpha\beta)$ from A to B.
3. Apply $(X^\beta Z^\alpha)^\dagger$ on B.

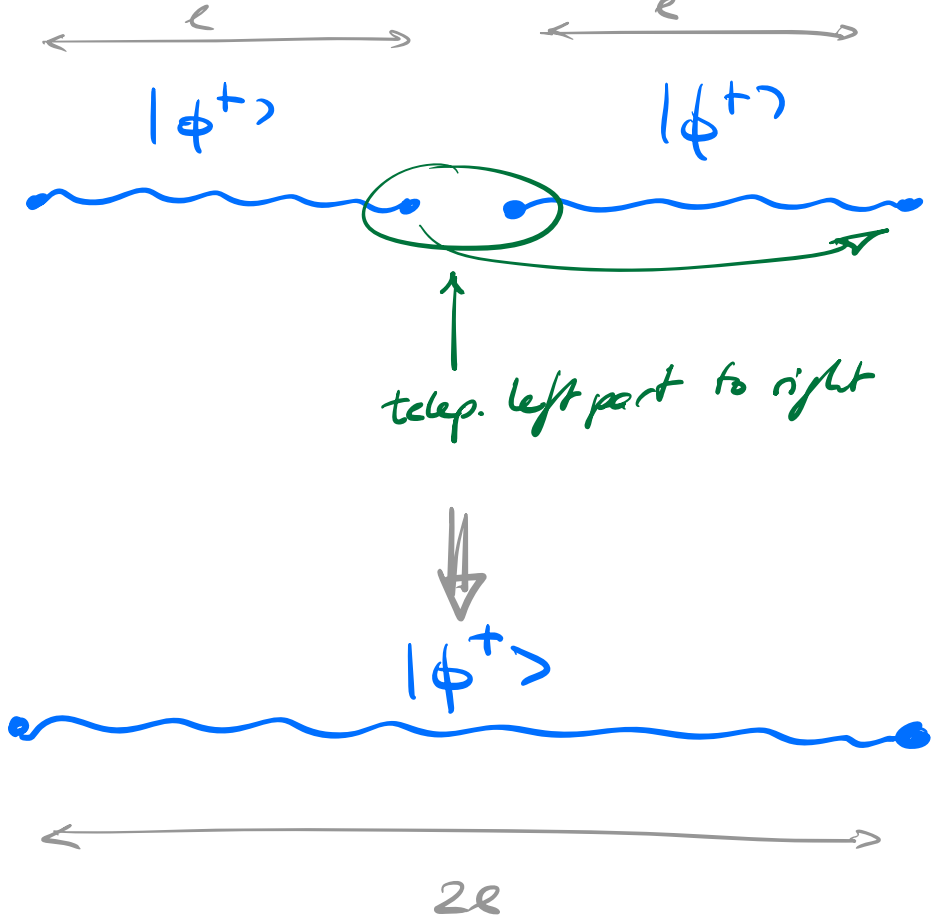
Can be straightforwardly generalized to $\mathbb{C}^d$.

One application of teleportation:

Quantum Repeaters

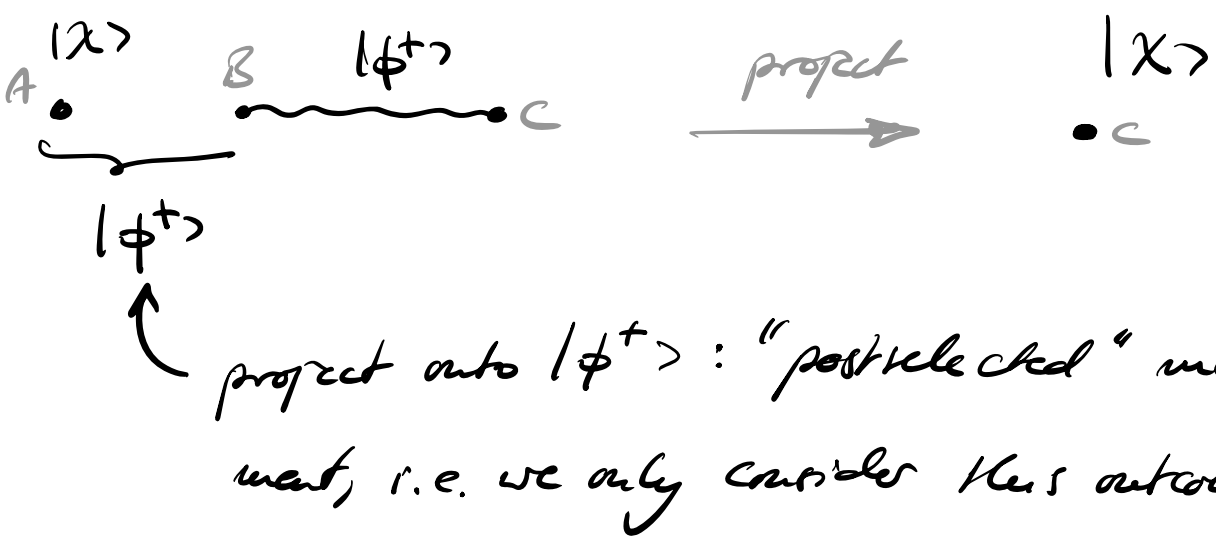
We can (reliably) create entanglement over distance ℓ → can we create entanglement over distance 2ℓ?

(E.g.: Photon loss at const. rate → prob. to send half of an ent. pair over dist. ℓ is $e^{-\ell/\xi}$.)



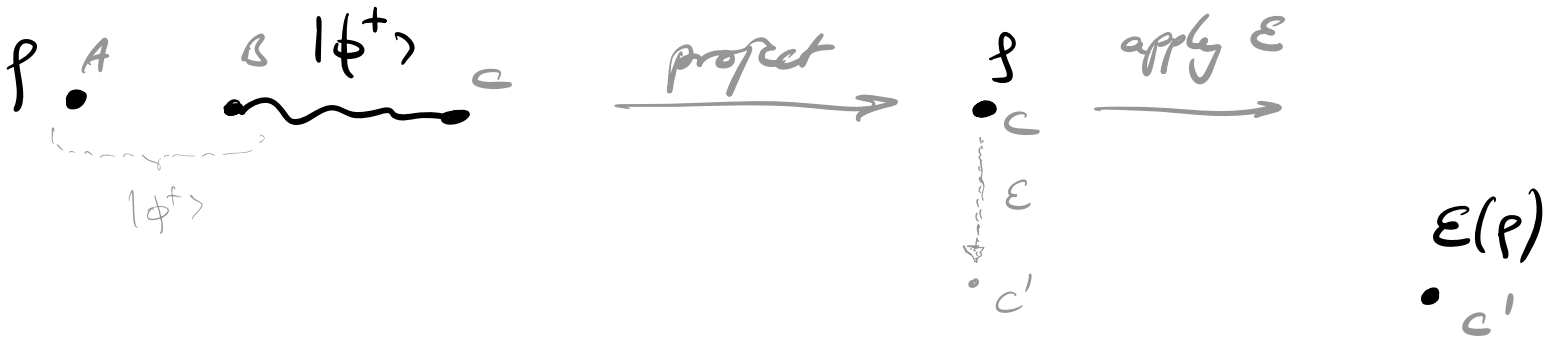
b) Relation between teleportation and the Choi-Jamiołkowski isomorphism

① Consider "postselected teleportation"

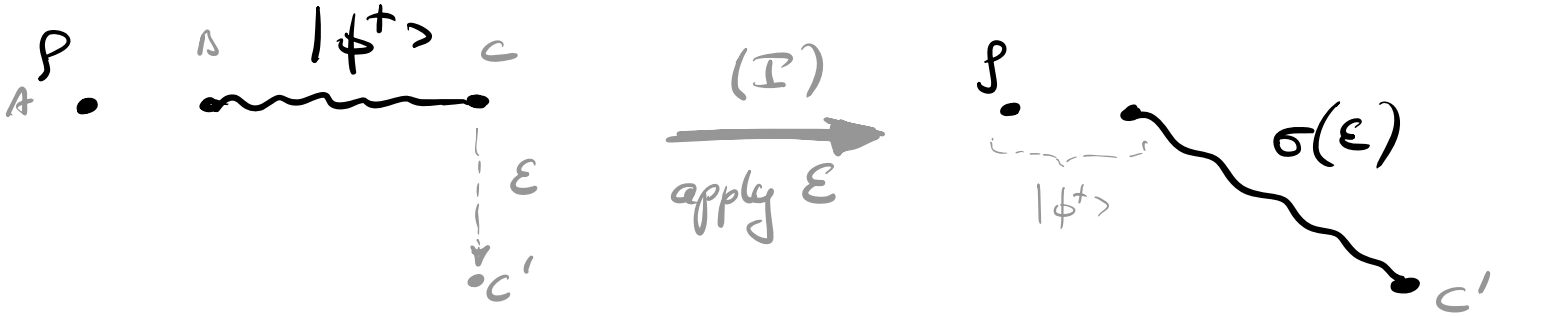


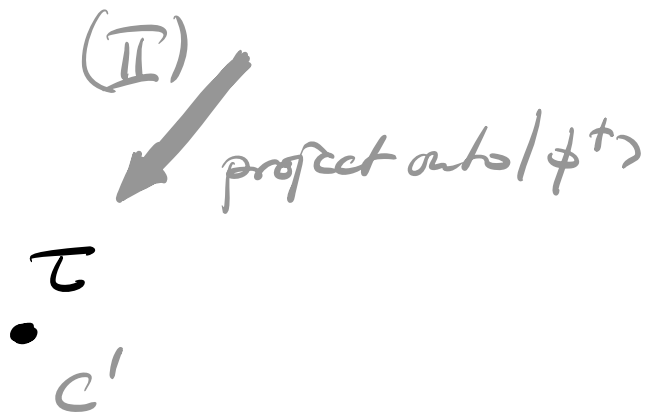
... so this is a complicated way of writing the identity map.

② Protocol for applying $\rho \mapsto \mathcal{E}(\rho)$:



③ Now <u>interchange the order of applying $\mathcal{E}$</u> and <u>projecting</u> — they commute (as they act on diff. systems), so this is the same map:





Since orders commute, this is the same as ②, i.e.

$$\tau = \mathcal{E}(\rho)\,!$$

Thus is the Choi-Jamiołkowski isomorphism (!):

(I) is the $\mathcal{E} \longmapsto \sigma$ map

("apply $\mathcal{E}$ to half a max. entangled state")

(II) is the $\sigma \longmapsto \mathcal{E}$ map

("teleport σ through the Choi state")

c) Dense coding

Have seen:

- shared entanglement + class. channel $\longrightarrow$ q. channel

1 ebit + 2 cbit $\longrightarrow$ 1 qubit

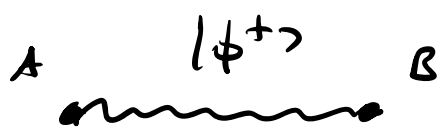
Can we do the converse? Use a quantum channel to transmit classical information?

Trivially possible by encoding $0 \to |0\rangle$, $1 \to |1\rangle$

1 qubit $\longrightarrow$ 1 cbit

Can we do better if we also share entanglement?

Dense coding (sometimes also "superdense coding"):



Idea: Encode two bits in $\{|\phi_{\alpha\beta}\rangle\}_{\alpha,\beta\in\{0,1\}}$ (an ONB)

(1) A & B share $|\phi^+\rangle$.

(2) A can encode two bits α,β locally:

$$|\phi_{\alpha\beta}\rangle_{AB} = (Z_A^\alpha X_A^\beta \otimes I)|\phi^+\rangle_{AB}$$

i.e., A applies $Z^\alpha X^\beta$ to her part of $|\phi^+\rangle$.

(3) A sends her part of the state to B via the quantum communication channel.

(4) B measures in Bell basis $\{|\phi_{\alpha\beta}\rangle\}$ and recovers α and β.

shared ent. + q. channel → class. channel

1 ebit + 1 qubit ⟶ 2 cbit

d) Optimality of teleportation & dense coding

We can use the teleportation & dense coding protocol mutually to argue that both are optimal in terms of communication cost.

To this end, assume shared ent. is free (i.e.: this is not part of our cost function).

i) Assume we can teleport with $r < 2$ bits of class. communication per qubit sent (i.e., there are protocols to send k_q qubits w/ k_c class. bits s.th., $\frac{k_c}{k_q} \longrightarrow r$).

Use this "hyper-teleportation" protocol to send quantum states in the (normal) dense coding prot.:

send $2n$ cbits

↓ dense coding

send n qubits

↓ "hyper-teleportation"

send rn cbits, $r < 2$

⇒ Can compress class. information (in the presence of entanglement).

⇒ Can iterate this to arbitrarily compress class. info – i.e., send n bits with $k \ll n$ bits – as long as we have free entanglement.

This is impossible! (Intuitively, can also be formalized.)

ii) Assume we can "hyper-dense-code" $2s > 2$ class. bits per qubit sent.

send $2s$ class. bits

↓

"hyper-dense coding"

send 1 qubit

teleportation

send 2 cbits

... and again, we can send 2 bits by only transmitting 2 bits, etc. ...

4. Entanglement conversion & quantification

a) Introduction and setup

Entanglement = those properties of a system which cannot be changed using **Local Operations and Classical Communication (LOCC)**.

(Note: This is typically the **definition** of entanglement.)

Question: When – and how – can we convert entangled states into another by LOCC?

Relevance:

- Different protocols might require different – "cheaper" or "more expensive" – entangled states.
- We can easily produce some entangled state but need a different one.
- Can be used to **quantify** entanglement via conversion rate, e.g. to some "gold standard"

– for instance, how many "e-bits" $|\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ are contained in a state? (And is there a meaning to "$|X\rangle$ contains 0.7 e-bits?")

Let us first consider pure states!

Known from Chapter I:

Same Schmidt coefficients $\Longleftrightarrow$ states can be converted by local unitaries (subclass of LOCC!)

Question: What if Schmidt coefficients are different?

Example: $|X\rangle = \sqrt{\frac{2}{3}}|00\rangle + \sqrt{\frac{1}{3}}|11\rangle$

$|\phi^+\rangle = \sqrt{\frac{1}{2}}|00\rangle + \sqrt{\frac{1}{2}}|11\rangle$

Question 1: Can we do $|\phi^+\rangle \xrightarrow{\text{LOCC}} |\chi\rangle$?

convert using LOCC.

A does POVM $\{\Pi_0, \Pi_1\}$,

$$\Pi_0 = \begin{pmatrix} \sqrt{2/3} & \\ & \sqrt{1/3} \end{pmatrix}; \quad \Pi_1 = \begin{pmatrix} \sqrt{1/3} & \\ & \sqrt{2/3} \end{pmatrix}.$$

Post-measurement states $|\tilde{\psi}_k\rangle = (\Pi_k \otimes I)|\phi^+\rangle$:

$$|\tilde{\psi}_0\rangle = \frac{1}{\sqrt{2}}\left(\sqrt{\frac{2}{3}}|00\rangle + \sqrt{\frac{1}{3}}|11\rangle\right)$$

$$|\tilde{\psi}_1\rangle = \frac{1}{\sqrt{2}}\left(\sqrt{\frac{1}{3}}|00\rangle + \sqrt{\frac{2}{3}}|11\rangle\right)$$

$\Rightarrow$

- $p_0 = 1/2$: $|\psi_0\rangle = \sqrt{\frac{2}{3}}|00\rangle + \sqrt{\frac{1}{3}}|11\rangle = |\chi\rangle$ ✓ SUCCESS!
- $p_1 = 1/2$: $|\psi_1\rangle = \sqrt{\frac{1}{3}}|00\rangle + \sqrt{\frac{2}{3}}|11\rangle$

→ wrong state, but right Schmidt coefficients → can be fixed by local unitaries!

(specifically, A & B need to apply $X \otimes X$).

Protocol for conversion $|\phi^+\rangle \xrightarrow{\text{LOCC}} |\chi\rangle$:

1. A does POVM $\{\Pi_0, \Pi_1\}$
2. A sends her outcome to B
3. If result is 1, both apply X.

$\Rightarrow$ Success probability $p = p_0 + p_1 = 1$!

Note: This is the best possible, since POVM cannot increase Schmidt rank $\rightarrow$ we cannot get more than one copy of $|\phi^+\rangle$ with any prob.!

Question 2: Can we do $|\chi\rangle \xrightarrow{\text{LOCC}} |\phi^+\rangle$?

A does POVM $\{\Pi_0, \Pi_1\}$,

$$\Pi_0 = \begin{pmatrix} \sqrt{1/2} & \\ & 1 \end{pmatrix}, \quad \Pi_1 = \begin{pmatrix} \sqrt{1/2} & \\ & 0 \end{pmatrix}$$

$$\rightarrow |\tilde{\phi_0}\rangle = \sqrt{1/3}\,|00\rangle + \sqrt{1/3}\,|11\rangle$$

$$|\tilde{\phi_1}\rangle = \sqrt{1/3}\,|00\rangle.$$

Outcome:

- $p_0 = \frac{2}{3}$: $|\psi_0\rangle = |\phi^+\rangle$ ✓ SUCCESS
- $p_1 = \frac{1}{3}$: $|\psi_1\rangle = |00\rangle$ ✗ FAILURE

(no ent. left → cannot salvage state!)

Protocol for $|X\rangle \xrightarrow{\text{LOCC}} |\phi^+\rangle$

1. A does POVM $\{\Pi_0, \Pi_1\}$.
2. A sends result to B.
3. If result is 0, they have obtained $|\phi^+\rangle$.
   Otherwise, they declare failure.

⟹ Success probability $p = p_0 = \frac{2}{3}$.

Note: Will see: This is indeed the best conversion rate possible.

Big drawback: Conversion is not reversible — we cannot use it to attach one number to an entangled state (i.e. quantify entanglement).

b) Single-copy conversion

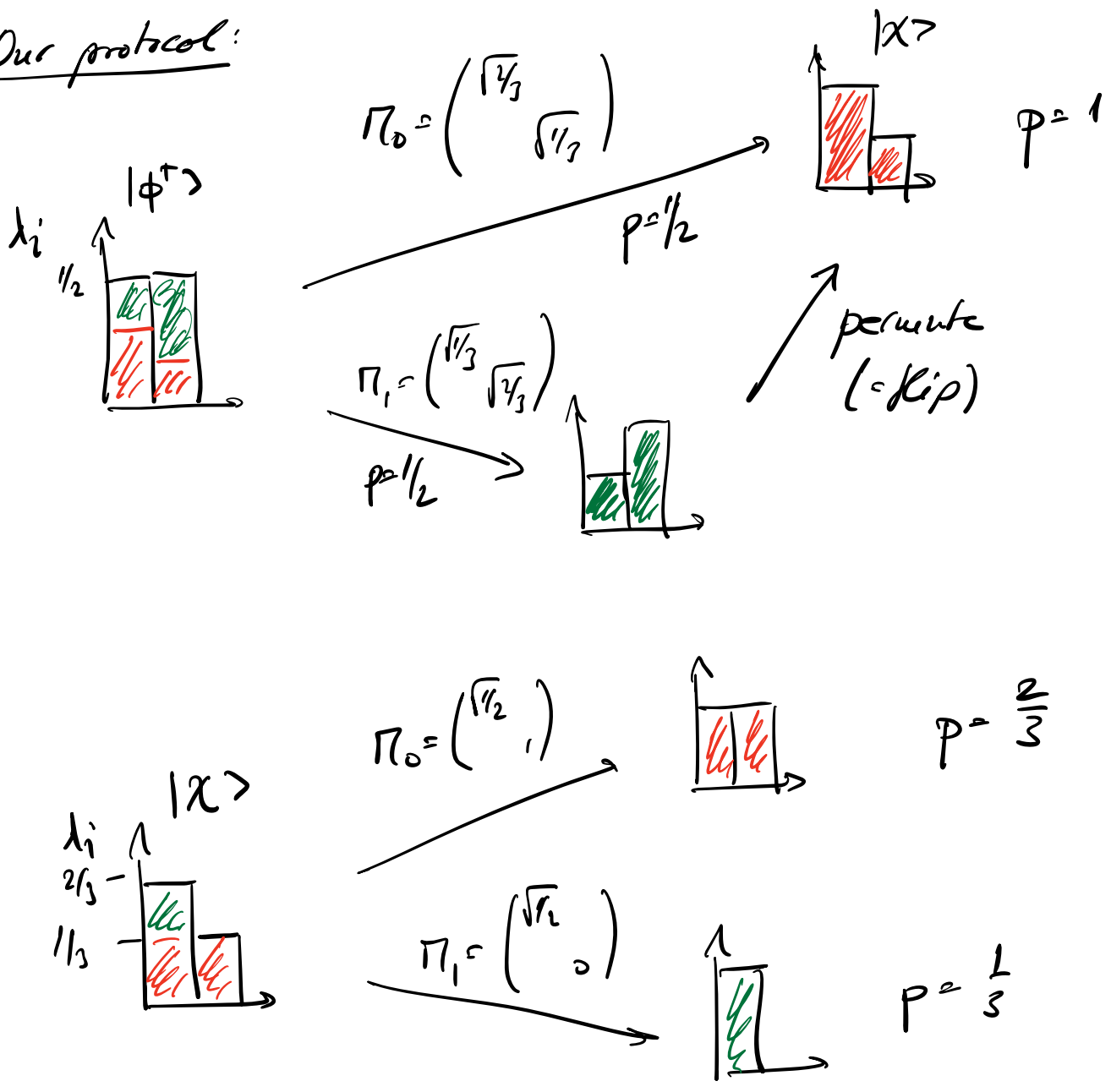
Can we generalize this approach?

Work in Schmidt basis:

$$|\psi\rangle = \sum_{i=1}^{d} \sqrt{\lambda_i}\, |i\rangle|i\rangle, \quad \lambda_1 \geq \lambda_2 \geq \dots$$

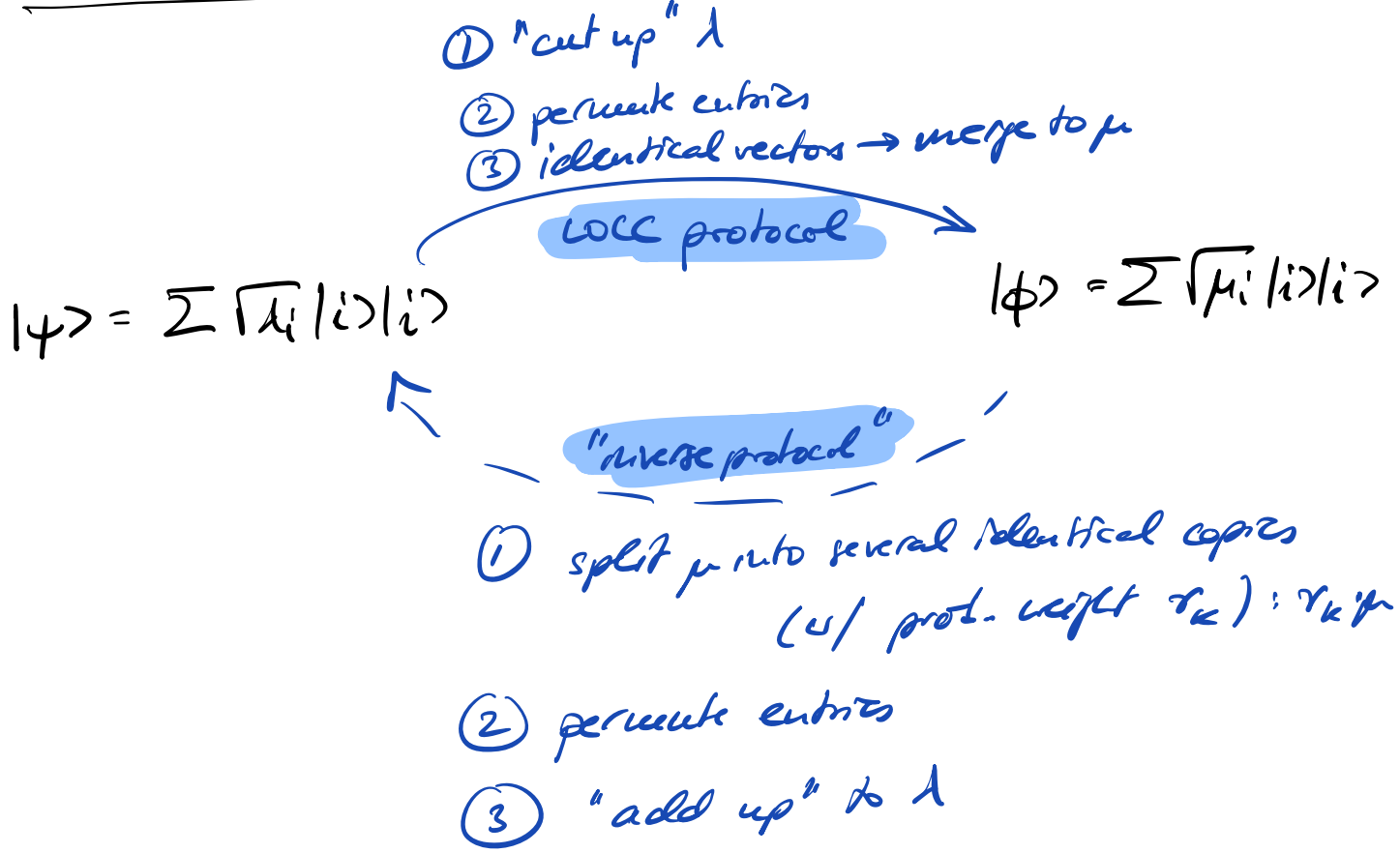
$\lambda = (\lambda_1, \dots, \lambda_d) \in \mathbb{R}^d$, $\sum \lambda_i = 1$: probability distribution

Our protocol:



When is this possible?

Deterministic conversion:



I.e.: Such a conversion is possible exactly if

$$\lambda = \sum r_k T_k \mu \qquad (*)$$

r_k: prob. distribution; T_k: Permutation

(Interpretation: Prob. distribution λ can be obtained by "scrambling" (= randomly permuting) μ.)

When does (*) hold?

Definition: (Majorization)

Let $\lambda, \mu \in \mathbb{R}^d$ be ordered probability vectors (i.e. $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0$, etc.)

We say μ majorizes λ (or λ is majorized by μ), written $\lambda \prec \mu$, iff

$$\sum_{i=1}^{k} \lambda_i \leq \sum_{i=1}^{k} \mu_i \quad \forall k = 1, \dots, d.$$

(Note: Majorization always defined for ordered vectors.)

Remarks:

- Majorization defines a partial order on the space of prob. distributions
- $\lambda \prec \mu$: λ "more disordered" than μ – in part., it has larger entropy! (Made rigorous by notion of "Schur concavity/convexity": for a concave/convex $f(x)$,

$$F(\lambda) = \sum f(\lambda_i) \text{ satisfies}$$

$$\lambda \prec \mu \Rightarrow F(\lambda) \underset{(\leq)}{\geq} F(\mu)$$

(≥: concave; ≤: convex)

Theorem:

$$\lambda \prec \mu \iff \exists r_k, \Pi_k : \quad \lambda = \sum_{k=1}^{s} r_k \Pi_k \mu$$

prob. distr. — permutations.

Proof: Homework.

Corollary: $|\psi\rangle = \sum \sqrt{\lambda_i}\,|i\rangle|i\rangle$ can be converted to $|\phi\rangle = \sum \sqrt{\mu_i}\,|i\rangle|i\rangle$ deterministically by LOCC if and only if* $\lambda \prec \mu$,

always: ordered vector!

* That is of course only true if we show that this class of protocols is the most general one which can be shown.

Remarks:

① One cannot improve by using POVMs which are not diagonal in the Schmidt basis.

(2) We have only considered LOCC protocols with a single round (A does POVM, sends result to B, A&B apply unitary). In principle LOCC protocols can have arbitrary # of rounds: A measures, sends result to B, B does meas. (which can dep. on information he received), sends result to A, A measures, ...

However: For pure states, there is no more powerful than single-copy protocol considered.

(Proof: Homework)

(3) The more general protocol would be

$$|\psi\rangle = \sum \sqrt{\lambda_i}\, |i\rangle|i\rangle$$

$$\xrightarrow{P_1} |\phi_1\rangle = \sum \sqrt{\mu_i^1}\, |i\rangle|i\rangle$$

$$\vdots$$

$$\xrightarrow{P_s} |\phi_s\rangle = \sum \sqrt{\mu_i^s}\, |i\rangle|i\rangle$$

This can be realized by LOCC if & only if

$$\lambda \prec \sum_{k=1}^{s} p_k \mu^k$$

(μ^k: ordered)

Example:

Optimal rate for $\left(\frac{1}{2}, \frac{1}{2}\right) \longleftrightarrow \left(\frac{2}{3}, \frac{1}{3}\right)$

↑ vectors with Schmidt coefs.

- $\left(\frac{1}{2}, \frac{1}{2}\right) \prec \left(\frac{2}{3}, \frac{1}{3}\right)$: prob = 1 ✓

- $\left(\frac{2}{3}, \frac{1}{3}\right) \prec \frac{2}{3}\left(\frac{1}{2}, \frac{1}{2}\right) + \frac{1}{3}(1,0) = \left(\frac{2}{3}, \frac{1}{3}\right)$

↑ max. possible value.

prob = 2/3 ✓

- we also see that Schmidt rank cannot be increased, e.g.: Can we do

$$|\phi^+\rangle \longrightarrow \left(\sqrt{\tfrac{2}{3}}|00\rangle + \sqrt{\tfrac{1}{3}}|11\rangle\right)^{\otimes 2} \; ?$$

$$\left(\tfrac{1}{2}, \tfrac{1}{2}, 0, 0\right) \not\prec p\left(\tfrac{4}{9}, \tfrac{2}{9}, \tfrac{2}{9}, \tfrac{1}{9}\right) + (1-p)\cdot \text{Shuff}$$

impossible!

c) Asymptotic protocols

Single-copy entanglement conversion:

not reversible!

E.g.: $|\phi^+\rangle \rightleftharpoons |\chi\rangle = \sqrt{\frac{2}{3}}|00\rangle + \sqrt{\frac{1}{3}}|11\rangle$

need 1 ebit for "$\rightarrow$",

get $\frac{1}{3}$ ebit from "$\leftarrow$".

→ Entanglement is lost

→ need at least two numbers to quantify entanglement:

ebits needed to build state

ebits extractable from state

Can we do better if we work with more copies?

$|\chi\rangle^{\otimes 2} \rightarrow p_2 |\phi^+\rangle^{\otimes 2} + p_1 |\phi^+\rangle^{\otimes 1}$?

$|\chi\rangle^{\otimes 3} \rightarrow p_3 |\phi^+\rangle^{\otimes 3} + p_2 |\phi^+\rangle^{\otimes 2} + \ldots$?

Average yield of max. ent. states $|\phi^+\rangle$

= "# of ebits per copy of $|\chi\rangle$":

$$\bar{P} = \frac{P_1 + 2P_2 + 3P_3 + \dots}{k}$$

$k \leftarrow$ # of copies.

E.g. for $|\chi\rangle^{\otimes 3}$:

$$\left(\frac{8}{27}, \frac{4}{27}, \frac{4}{27}, \frac{4}{27}, \frac{2}{27}, \frac{2}{27}, \frac{2}{27}, \frac{1}{27}\right) \prec$$

$$P_3\left(\tfrac{1}{8}, \tfrac{1}{8}, \tfrac{1}{8}, \tfrac{1}{8}, \tfrac{1}{8}, \tfrac{1}{8}, \tfrac{1}{8}, \tfrac{1}{8}\right) +$$

$$P_2\left(\tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}, 0, 0, 0, 0\right) +$$

$$P_1\left(\tfrac{1}{2}, \tfrac{1}{2}, 0, 0, 0, 0, 0, 0\right) +$$

$$P_0\,(1, 0, 0, 0, 0, 0, 0, 0)$$

$$\| \quad 1-(P_1+P_2+P_3)$$

$P_3 = 8/27$, $P_2 = 16/27$, $P_1 = 0$, $P_0 = 3/27$,

gives a valid solution.

$$\bar{p} = \frac{3 \cdot \frac{8}{27} + 2 \cdot \frac{11}{27}}{3} = \frac{56}{81} > \frac{2}{3}!$$

→ Improvement over single-copy protocols!

However: Finite # of copies will probably not allow for reversible conversion, as Schmidt rank cannot increase.

How good can we get by using $N \to \infty$ copies?

Requirements for asymptotic protocols

- Convert $|\phi^+\rangle^{\otimes n} \rightleftarrows |X\rangle^{\otimes N}$ with a rate $R = \lim \frac{n}{N}$ as $N, n \to \infty$

  ↑ we aim for best rate R:
  large R for "←", small R for "→"

- Need not be deterministic:

  Success probability $p \to 1$ as $N, n \to \infty$

- Need not be perfect:

Require that distance δ from correct state goes to zero, $\delta \to 0$, as $N, n \to \infty$.

(Note: We can afford these imperfections since asymptotically, they vanish: This does not make sense in finite-copy scenario.

We also need these imperfections since we cannot increase the Schmidt rank.)

How can we measure error δ?

Definition (Fidelity): For two states $|\psi\rangle$ and $|\phi\rangle$,

$F = |\langle\psi|\phi\rangle|^2$ is called the *fidelity* of $|\psi\rangle$ and $|\phi\rangle$.

Lemma: $\delta := 1 - F$ bounds the error on any expectation value,

$$|\langle\psi|O|\psi\rangle - \langle\phi|O|\phi\rangle| \leq 2\sqrt{\delta}\,\|O\|_\infty,$$

with $\|O\|_\infty = \sup \frac{\|O|\psi\rangle\|}{\||\psi\rangle\|}$ the operator norm.

That is: δ bounds how well we can (or cannot) distinguish $|\phi\rangle$ and $|\psi\rangle$ by any physical test!

Proof: → Homework.

Structure of $|\chi\rangle^{\otimes N}$:

Let us now consider some state

$$|\chi\rangle = \sum_{x=1}^{d} \sqrt{p(x)}\ |x\rangle_A |x\rangle_B .$$

Then, $|\chi\rangle^{\otimes N} = \sum_{x_1,\dots,x_N} \sqrt{p(x_1)\cdot\dots\cdot p(x_N)}\ |x_1,\dots,x_N\rangle_A |x_1,\dots,x_N\rangle_B$,

i.e., the probabilities $p(x_1,\dots,x_N)$ associated to a configuration $x_1,\dots,x_N$ describe independently & identically distributed (iid) random variables with prob. $p(x_i)$.

If we measure $|x_1,\dots,x_N\rangle_A$ in the computational basis (i.e., sample $x_1,\dots,x_N$ from $p(x)$), what do we expect?

"Typical" output: Expect output x $N \cdot p(x)$ times.

$$\Longrightarrow p(x_1, \dots, x_N) = p(x_1) \cdot \dots \cdot p(x_N) \approx p(1)^{Np(1)} \cdot \dots \cdot p(d)^{Np(d)}$$

$$\Longrightarrow -\log p(x_1, \dots, x_N) \approx N \cdot \underbrace{\left(- \sum_{x=1}^{d} p(x) \log p(x) \right)}_{=: H(p) : \text{Shannon entropy of } p.}$$

our logs are base 2!

$\Longrightarrow$ we typically expect to see sequences $x_1, \dots, x_N$ for which

$$p(x_1, \dots, x_N) \approx 2^{-NH(p)},$$

and there are $\approx 2^{NH(p)}$ such typical sequences.

More formally, this is defined as follows:

Definition (ε-typical sequences):

For an i.i.d-variable x, we say that $x_1, \dots, x_N$ is an ε-typical sequence if

$$2^{-N(H(p)+\varepsilon)} \leq p(x_1, \dots, x_N) \leq 2^{-N(H(p)-\varepsilon)}$$

We denote the set of ε-typical sequences by $T(N, \varepsilon)$.

Theorem:

$\forall \varepsilon > 0 \; \forall \delta > 0 \; \exists N_0 \; \forall N \geq N_0$ such that

1) a random sequence $x_1, \dots, x_N$ of length N is ε-typical w/ prob. $\geq 1-\delta$.

2) $(1-\delta)\, 2^{N(H(p)-\varepsilon)} \leq |T(N, \varepsilon)| \leq 2^{N(H(p)+\varepsilon)}$

Proof idea:

Interpret $(-\log p(x_i))_i$ as an i.i.d. random variable, with probability $p(x_i)$ for outcome i.

Then $p(x_1, \dots, x_N) = \frac{1}{N} \sum_{i=1}^{N} \left(-\log p(x_i)\right)$

and $H(p) = \sum_i p_i \left(-\log p(x_i)\right)$.

1) Then follows from the law of large numbers,

and 2) from summing (1) over all typ. sequences.

□

In words:

ε-typical sequence: $\Leftrightarrow$ $\frac{\log p(x_1, \dots, x_N)}{N}$ ε-close to $H(p)$

Asymptotically, a sequence is ε-typical w/ prob $\to 1$,

and there are $\sim 2^{NH(p)}$ ε-typical sequences.

Note: Typical sequences are an important concept in classical information theory ($\to$ data compression etc.!)

Application of typicality to entanglement conversion:

Protocol for $|\phi^+\rangle^{\otimes n} \longrightarrow |\chi\rangle^{\otimes N}$:

$$|\chi\rangle^{\otimes N} = \sum_{x_1,\dots,x_N} \sqrt{p(x_1)\cdots p(x_N)}\; |x_1,\dots,x_N\rangle_A\, |x_1\dots x_N\rangle_B$$

Fix some $\varepsilon > 0$ and $\delta > 0$, and a matching N_0, and $N \geq N_0$. Let

$$|\tilde{\vartheta}_N\rangle := \sum_{x_1,\dots,x_N \in T(N,\varepsilon)} \sqrt{p(x_1)\cdots p(x_N)}\; |x_1,\dots,x_N\rangle |x_1,\dots,x_N\rangle,$$

$$|\vartheta_N\rangle := \frac{|\tilde{\vartheta}_N\rangle}{\sqrt{\langle\tilde{\vartheta}_N|\tilde{\vartheta}_N\rangle}}$$

Then:

$$\underbrace{\langle\vartheta_N|\left(|\chi\rangle^{\otimes N}\right)}_{\text{Fidelity } F} = \frac{\sum_{T(N,\varepsilon)} p(x_1,\dots,x_N)}{\sqrt{\sum_{T(N,\varepsilon)} p(x_1,\dots,x_N)}} \geq \sqrt{1-\delta} \geq 1-\delta$$

$\geq 1-\delta$ for $N \geq N_0$

$\delta = 1-F$: error.

and $|T(N,\varepsilon)| \leq 2^{N(H(p)+\varepsilon)}$

Protocol:

A prepares $|\vartheta_N\rangle$ locally & teleports Bob's part to Bob

$\Rightarrow$ uses $m = \log |T(N,\varepsilon)| \leq N(H(\rho)+\varepsilon)$ copies of $|\phi^+\rangle$.

Resulting state $|\vartheta_N\rangle$ is δ-close to $|\chi\rangle^{\otimes N}$.

$$\lim \frac{m}{N} = H(\rho) + \varepsilon \quad \text{"entanglement dilution rate"}$$

Since any $\varepsilon > 0$ admissible: Can achieve

asymptotic rate $R_{\text{dilute}} = H(\rho)$ for ent. dilution.

(Alternatively: use majorization, same result).

Protocol for $|\chi\rangle^{\otimes N} \longrightarrow |\phi^+\rangle^{\otimes m}$:

$$|\chi\rangle^{\otimes N} = \sum_{x_1,\dots,x_N} \sqrt{p(x_1)\cdots p(x_N)}\; |x_1,\dots,x_N\rangle_A\, |x_1\dots x_N\rangle_B$$

A projects onto ε-typ. subspace $\longrightarrow$ obtains $|\vartheta_N\rangle$.

(i.e.: POVM $\{\Pi_0 = \Pi_{\varepsilon\text{-typ}},\ \Pi_1 = \mathbb{I} - \Pi_0\}$)

$|\tilde{\vartheta}_N\rangle$: largest Schmidt coeff. $\leq 2^{-N(H(\rho)-\varepsilon)}$

$\Rightarrow$ $|\vartheta_N\rangle$: largest Schmidt coeff.

$$\leq \frac{1}{1-\delta} 2^{-N(H(\rho)-\varepsilon)}$$

Choose M s.t. $\frac{1}{1-\delta} 2^{-N(H(\rho)-\varepsilon)} \leq 2^{-M}$

$\Rightarrow$ $(\underbrace{2^{-M}, 2^{-M}, \ldots, 2^{-M}}_{2^M \text{ times}}, 0, \ldots 0) \succ$ (Schmidt coeff. of $|\vartheta_N\rangle$)

$\Rightarrow$ can convert $|\vartheta_N\rangle$ to $|\phi^+\rangle^{\otimes M}$ by LOCC.

Protocol works for any $M \leq N(H(\rho)-\varepsilon) + \log(1-\delta)$

$\Rightarrow$ Rate $R = \lim \frac{M}{N} \longrightarrow H(\rho) - \varepsilon \quad \forall \varepsilon > 0.$

$\Rightarrow$ Asymptotic rate $R_{dist} = H(\rho)$ for entanglement distillation.

We find: Asymptotically (i.e., $N, M \to \infty$),

distillation rate = dilution rate = $H(\rho)$

Is this optimal?

→ Yes! Otherwise, we could "go in circles" and increase # of $|\phi^+\rangle$ in every iteration by LOCC! ↯

Note: Instead of the Shannon entropy $H(p)$, we typically use the

$$\boxed{\text{von Neumann entropy} \quad S(\rho) = -\operatorname{tr}(\rho \log \rho)}$$

- since $\log\rho$ is defined in the eigenbasis, we have

$$-\operatorname{tr}\rho\log\rho = -\sum p_i \log p_i \quad \text{with } (p_i) = \operatorname{eig}(\rho),$$

i.e., $H(p) = S(\operatorname{tr}_B |\psi\rangle\langle\psi|) = S(\operatorname{tr}_A |\psi\rangle\langle\psi|)$.

Note: This protocol allows us to go between any two states $|\psi\rangle^{\otimes N}$ and $|\phi\rangle^{\otimes M}$ asymptotically reversibly, provided

$$\boxed{N \cdot S(\operatorname{tr}_B |\psi\rangle\langle\psi|) = M \cdot S(\operatorname{tr}_B |\phi\rangle\langle\phi|)}$$

(by going via $|\phi^+\rangle^{\otimes K}$)

$S(\mathrm{tr}_B|\psi\rangle\langle\psi|) =: E(|\psi\rangle)$ can thus be seen as a unique measure of entanglement (in an asymptotic LOCC setting).

Key result: The entropy of entanglement

$$E(|\psi\rangle) = S(\mathrm{tr}_A|\psi\rangle\langle\psi|) = S(\mathrm{tr}_B|\psi\rangle\langle\psi|)$$

uniquely quantifies the amount of entanglement in a pure bipartite state in an asymptotic setting.

5. Entanglement of mixed states

a) Introduction

When is a mixed bipartite state ρ_{AB} entangled?

Different possible definitions:

(i) If ρ_{AB} cannot be created by LOCC.

(ii) If we can extract (bits) $|\phi^+\rangle$ from ρ_{AB}.

(iii) If it helps us to do some task better in an LOCC setting (if is a "resource").

... any of those could be in a finite-copy or asymptotic setting!

Clearly, (ii) $\geq$ (iii) $\geq$ (i)

↑ "$\geq$": is a stronger condition, i.e. satisfied by less states.

We use the weakest notion (i) to define entangled states.

Definition:

A bipartite state ρ_{AB} is called **separable** if it can be written as

$$\rho_{AB} = \sum_i p_i \rho_i^A \otimes \rho_i^B \qquad (\otimes)$$

"separable state"

for some $p_i \geq 0$, $\sum p_i = 1$;

$\rho_i^A \geq 0$, $\rho_i^B \geq 0$; $\mathrm{tr}\rho_i^A = \mathrm{tr}\rho_i^B = 1$

— i.e., ρ_{AB} can be prepared by LOCC.

If ρ_{AB} is not separable — i.e., it has no decomposition of the form $(\otimes)$ — it is called **entangled**.

Given a state ρ_{AB}, how can we tell if it is entangled?

Problem: Given ρ_{AB}, need to check **all** decompositions

$$\rho_{AB} = \sum_i p_i \rho_i^{AB} = \sum_i p_i \rho_i^A \otimes \rho_i^B$$

(or $\rho_{AB} = \sum p_i |\psi_i^{AB}\rangle\langle\psi_i^{AB}|$) to find if there is a separable one. (Ambiguity in ensemble decomposition is an isometry, i.e., we need to optimize over isometries!)

→ Difficult!

(In fact, the general problem has been shown to be NP-hard in the dimension of the space.)

→ Need partial solutions, e.g. ways to certify a given state is entangled.

b) Entanglement witnesses

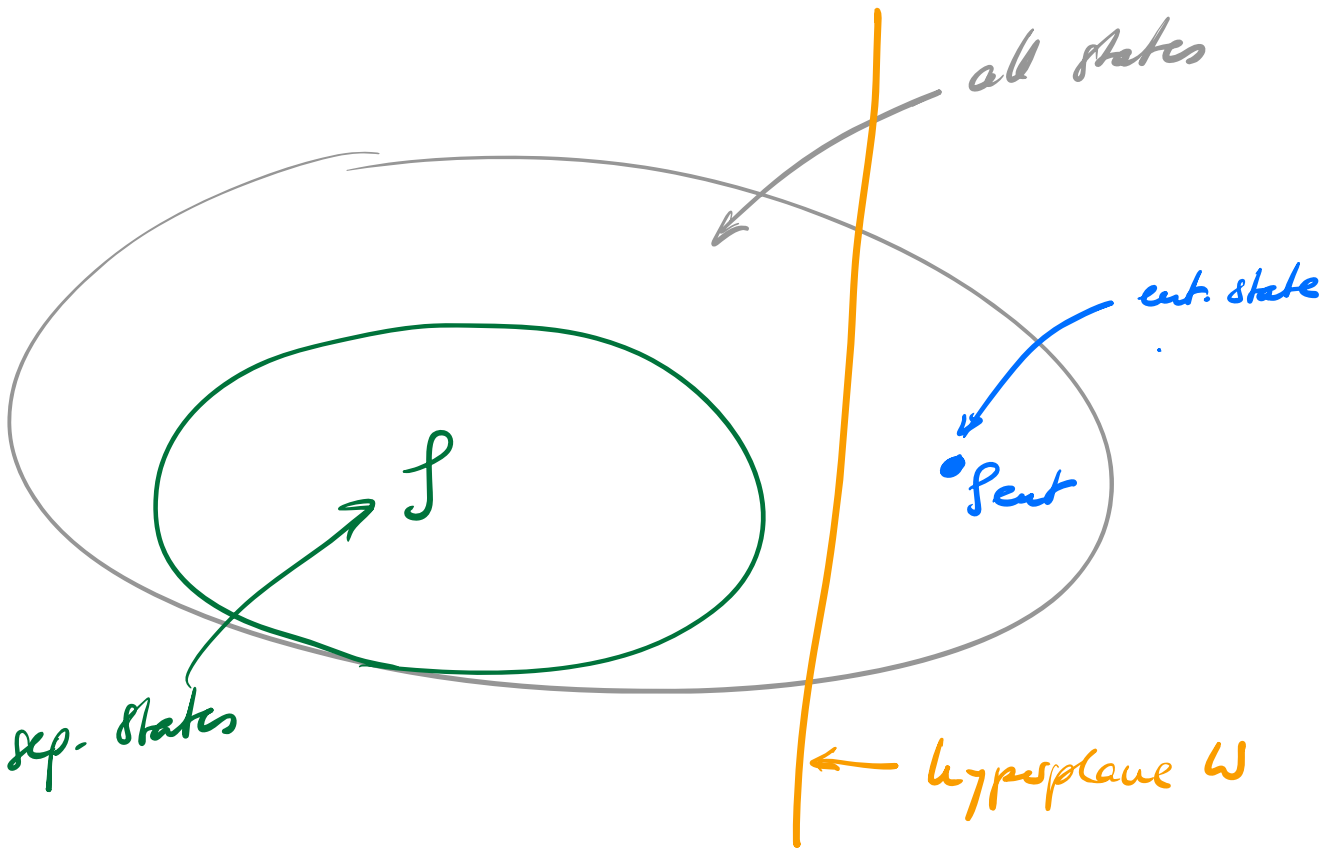
Structure of the set $\mathcal{S}$ of separable states:

$$\text{let } \rho = \sum_{i=1}^{N} R_i \rho_i^A \otimes \rho_i^B \in \mathcal{S}_j \quad \sigma = \sum_{j=1}^{M} q_j \sigma_j^A \otimes \sigma_j^B \in \mathcal{S}$$

$\Rightarrow$ for $\lambda \in [0;1]$,

$$\lambda\rho + (1-\lambda)\sigma = \lambda p_1 \rho_1^A \otimes \rho_1^B + \dots + \lambda p_N \rho_N^A \otimes \rho_N^B + \dots$$
$$+ (1-\lambda) q_1 \sigma_1^A \otimes \sigma_1^B + \dots + (1-\lambda) q_M \sigma_M^A \otimes \sigma_M^B \in \mathcal{S}$$

$\Rightarrow$ the set $\mathcal{S}$ of separable states forms a convex set



For any state $\rho_{ent} \notin \mathcal{S}$, we can find a hyperplane which separates ρ_{ent} from $\mathcal{S}$.

More generally, we can construct hyperplanes

s.th. all points on one side of the plane are entangled (but not the other way).

Any hyperplane is of the form $\mathrm{tr}[X\rho] + c = 0$,

i.e., $\mathrm{tr}[\underbrace{(X + cI)}_{=:W}\rho] = \mathrm{tr}[W\rho] = 0.$

We can choose $W = W^\dagger$ as we work in the space of hermitian matrices.

Then:

$\mathrm{tr}(\rho W) > 0$: ρ left of hyperplane W

$\mathrm{tr}(\rho W) < 0$: ρ right of hyperplane W

I.e. for a hyperplane as above:

ρ separable $\Rightarrow$ $\mathrm{tr}(W\rho) \geq 0$

and thus

$\mathrm{tr}(W\rho) < 0 \Rightarrow \rho$ entangled.

Definition (entanglement witness):

An operator $W = W^\dagger$ such that

$$\rho \text{ separable} \Rightarrow \mathrm{tr}(W\rho) \geq 0$$

is called an entanglement witness.

Observation: Given an entanglement witness W,

$$\mathrm{tr}(\rho W) < 0 \Rightarrow \rho \text{ entangled.}$$

Notes:

- Key point: Need some way to prove that $\mathrm{tr}(W\rho) \geq 0 \quad \forall \rho \in S$!
- Any given witness will only detect some entangled states.
- S is a convex set $\Rightarrow$ S is fully specified by all its tangent planes $\Rightarrow$ there exists a witness for any entangled states.

- Witnesses are linear operators ⇒ they can be experimentally measured and can (and are) been long used to identify entanglement in experiments.

Example:

$$W = \mathbb{F} := \sum_{i,j=1}^{d} |i,j\rangle\langle j,i| \quad \text{(the "flip" or swap)}$$

Is it a witness?

let $\rho_{sep} = \sum p_i \rho_i^A \otimes \rho_i^B$:

$$\mathrm{tr}(W \rho_{sep}) = \sum p_i \, \mathrm{tr}\left(\mathbb{F}\left(\rho_i^A \otimes \rho_i^B\right)\right) = \dots$$

Use:

(1) $\mathrm{tr}[\mathbb{F}\cdot(A\otimes B)] = \sum_{ij} \mathrm{tr}[|i\rangle\langle j| A \otimes B]$

$$= \sum_{ij} \langle j,i|(A\otimes B)|i,j\rangle = \sum_{ij} \langle j|A|i\rangle\langle i|B|j\rangle$$

$$= \mathrm{tr}(AB) \quad \text{(the "magic formula")},$$

(2) $P, Q \geq 0 \Longrightarrow \mathrm{tr}(PQ) \geq 0$

Proof: $P = \sum p_i |\phi_i\rangle\langle\phi_i|$, $p_i \geq 0$.

$$\mathrm{tr}(PQ) = \sum_i \mathrm{tr}\left(p_i |\phi_i\rangle\langle\phi_i| Q\right)$$

$$= \sum_i p_i \underbrace{\langle\phi_i|Q|\phi_i\rangle}_{\geq 0} \geq 0.$$

$$\ldots = \sum p_i \underbrace{\mathrm{tr}\left(\rho_i^{A} \rho_i^{B}\right)}_{\geq 0} \geq 0.$$

$\Longrightarrow W$ is entanglement witness!

Which ent. states does W detect? — Those with dominant anti-symmetric component!

E.g. $d=2$, $|\psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$:

$$F|\psi^-\rangle = -|\psi^-\rangle \Longrightarrow \langle\psi^-|F|\psi^-\rangle = -1 < 0 !$$

while for e.g. $|\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$,

$$F|\phi^+\rangle = |\phi^+\rangle \longrightarrow \langle\phi^+|F|\phi^+\rangle = +1 > 0 \;\lightning$$

What about mixed states?

E.g. for $d=2$:

$$\rho = \lambda |\psi^-\rangle\langle\psi^-| + (1-\lambda)\frac{\mathbb{I}}{4} \quad ; \quad \lambda \in [-\tfrac{1}{3}, 1]$$

"Werner state"

$$\mathrm{tr}(F\rho) = \lambda \langle\psi^-|F|\psi^-\rangle + \frac{(1-\lambda)}{4}\underbrace{\mathrm{tr}(\mathbb{I}\cdot F)}_{= d = 2}$$

$$= -\lambda + \frac{1-\lambda}{2} = \frac{1}{2}(1-3\lambda)$$

$\rightarrow$ for $\lambda > 1/3$, ρ is entangled

Definition: A witness W is called optimal if there exist ρ separable such that $\mathrm{tr}[W\rho] = 0$.

(i.e., it touches the convex set & cannot be moved closer; otherwise, we could move it in parallel and get a strictly better witness.)

Is $W = F$ optimal?

Yes, e.g. $\rho = |0\rangle\langle 0| \otimes |1\rangle\langle 1|$

$\Rightarrow \operatorname{tr}(\rho F) = 0.$

Other witnesses: E.g. $W = I - d|\Omega\rangle\langle\Omega|$,

$|\Omega\rangle = \frac{1}{\sqrt{d}} \sum_{i=1}^{d} |i,i\rangle$ ⟶ Homework!

c) Positive maps and the PPT criterion

Reminder: A superoperator $\Lambda : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H})$ is called positive if $\rho \geq 0 \Rightarrow \Lambda(\rho) \geq 0$.

Usually – i.e. for physical maps – we want Λ in addition to be completely positive, i.e.

$\rho_{AB} \geq 0 \Rightarrow (\Lambda_A \otimes I_B)(\rho_{AB}) \geq 0.$

But now we will be interested in positive but not completely positive maps!

Why?

Consider $\rho_{sep} = \sum_i p_i \rho_i^A \otimes \rho_i^B$. Then,

$$(\Lambda \otimes I)(\rho_{sep}) = \sum_i p_i \underbrace{\Lambda(\rho_i^A)}_{=: \tilde{\rho}_i^A \geq 0 \text{ (Λ positive!)}} \otimes \rho_i^B$$

$$= \sum_i \underbrace{p_i}_{\geq 0} \underbrace{\tilde{\rho}_i^A}_{\geq 0} \otimes \underbrace{\rho_i^B}_{\geq 0} \geq 0$$

and thus:

Theorem: Let $\Lambda : B(\mathcal{H}) \to B(\mathcal{H})$ be a positive map.

Then, $(\Lambda \otimes I)(\rho_{AB}) \not\geq 0 \Rightarrow \rho_{AB}$ entangled.

↑ not positive semi-definite, i.e. has negative eigenvalues.

Most important example:

$$\Lambda(\rho) = \rho^T \quad \text{(the \underline{transpose map})}$$

$(\Lambda \otimes I)(\rho) =: \rho^{T_A}$ "partial transpose"

(cf. also Section I.5)

E.g.: $|\Omega\rangle = \frac{1}{\sqrt{d}} \sum_{i=1}^{d} |i,i\rangle$

$$\Rightarrow \left(|\Omega\rangle\langle\Omega|\right)^{T_A} = \frac{1}{d} \left(\sum |i,i\rangle\langle j,j|\right)^{T_A}$$

$$= \frac{1}{d} \sum |j,i\rangle\langle i,j|$$

$= \mathbb{F}$!

Not positive – entangled states have neg. eigenvalues,

i.e.: $|A\rangle = \frac{1}{\sqrt{2}} \left(|ij\rangle - |ji\rangle\right)$, $i \neq j$:

$$\langle A | \left(|\Omega\rangle\langle\Omega|\right)^{T_A} | A \rangle = \frac{1}{d} (-1) = -\frac{1}{d}.$$

Mixed states:

E.g. $\rho = \lambda |\Omega\rangle\langle\Omega| + (1-\lambda) \frac{I}{d^2}$, $\frac{-1}{d^2-1} \leq \lambda \leq 1$:

"isotropic state"

For instance for $d=2$:

$$\rho = \lambda \begin{pmatrix} \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{pmatrix} + (1-\lambda) \begin{pmatrix} \frac{1}{4} & & & \\ & \frac{1}{4} & & \\ & & \frac{1}{4} & \\ & & & \frac{1}{4} \end{pmatrix} = \begin{pmatrix} \frac{1+\lambda}{4} & & & \frac{1}{2} \\ & \frac{1-\lambda}{4} & & \\ & & \frac{1-\lambda}{4} & \\ \frac{1}{2} & & & \frac{1+\lambda}{4} \end{pmatrix}$$

(T_A: $\frac{1}{2} \to \frac{1-\lambda}{4}$, $\frac{1}{2} \to \frac{1-\lambda}{4}$)

$$\Rightarrow \rho^{T_A} = \begin{pmatrix} \frac{1+\lambda}{4} & & & \\ & \frac{1-\lambda}{4} & \frac{\lambda}{2} & \\ & \frac{\lambda}{2} & \frac{1-\lambda}{4} & \\ & & & \frac{1+\lambda}{4} \end{pmatrix}$$

positive iff $\left(\frac{1-\lambda}{4}\right)^2 \geq \left(\frac{\lambda}{2}\right)^2$ & $\frac{1+\lambda}{4} \geq 0$

$\Leftrightarrow -1 \leq \lambda \leq 1/3$

$\Rightarrow \rho^{T_A} \geq 0$ iff $\lambda \in \left[\frac{-1}{d^2-1} ; \frac{1}{3}\right]$

$\Rightarrow$ For $\lambda \in \left(\frac{1}{3}; 1\right]$, ρ is entangled &

Corollary (PPT criteria):

$$\rho^{T_A} \not\geq 0 \Rightarrow \rho \text{ entangled.}$$

This is called the **PPT (positive partial transpose) criteria**, or also **NPT criteria.** (not positive...).

Note: PPT criterion – and generally positive maps – are invariant under local unitaries on B ⟶ PPT detects all maximally ent. states (and in fact all states $|\chi\rangle$ w/ full Schmidt rank, as $|\chi\rangle = (I_A \otimes \pi_B)|\Omega\rangle$, π_B invertible).

That is: Positive maps are stronger than witnesses (will formalize this later) – but they cannot be measured.

In fact:

Theorem: The PPT criteria detects all entangled states in $d_A \times d_B = 2\times 2$ and 3×2 dimensions,

$$\rho \text{ entangled} \iff \rho^{T_A} \ngeq 0$$

(Not proven here.)

Counterexamples exist in $d_A \times d_B = 3\times 3$ and 2×4, i.e. states ρ with $\rho^{T_A} \geq 0$ which are entangled.

("PPT bound entangled states", cf. later.)

Other example of a positive but not CP map:

$$\Lambda(\rho) = \mathrm{tr}(\rho)\, I - \rho$$

$$(\Lambda \otimes I)(\rho_{AB}) = (I \otimes \mathrm{tr}_A \rho_{AB}) - \rho_{AB}.$$

$$I \otimes \mathrm{tr}_A \rho_{AB} - \rho_{AB} \neq 0 \Rightarrow \rho \text{ entangled.}$$

The "reduction criterion" for entanglement:

$$I \otimes \mathrm{tr}_A \rho \neq \rho \qquad (\rightarrow \text{Homework}).$$

d) Relation between witnesses and positive maps:

For each witness W, there is a positive map which detects all entangled states the witness detects — and in fact more. More precisely:

Theorem: There is an isomorphism between witnesses W on $\mathbb{C}^d \times \mathbb{C}^d$ and positive maps $\Lambda : B(\mathbb{C}^d) \to B(\mathbb{C}^d)$.

Moreover, if $\mathrm{tr}(W\rho) < 0 \Rightarrow (\Lambda \otimes I)(\rho) \not\geq 0$.

Proof:

Reminder (Choi-Jam. isomorphism):

CP map $\mathcal{E}$ on A — state σ on AB $(d_A = d_B)$

$$\mathcal{E} \longmapsto \sigma = (\mathcal{E} \otimes I)(|\Omega\rangle\langle\Omega|)$$

$$\mathcal{E}(\rho) = d\,\mathrm{tr}_B\left(\sigma\,(I \otimes \rho^T)\right) \longleftarrow \sigma$$

Isomorphism $\mathcal{E} \rightleftarrows \sigma$ applies also outside of CP maps and $\sigma \geq 0$.

Proof idea: Interpret W^T as the "Choi state" of a map

1. (Since $W \not\geq 0$ — otherwise $\mathrm{tr}(W\rho) \geq 0$ $\forall \rho$! — Λ is not CP.)

i.e.: $\Lambda(X) := d\,\mathrm{tr}_B\left(W^T(\mathbb{I}\otimes X^T)\right)$

$= d\,\mathrm{tr}_B\left(W(\mathbb{I}\otimes X)\right)^T.$

Need to show.

i) W witness $\Rightarrow$ $\Lambda(X)$ positive map.

ii) $\Lambda(X)$ positive map $\Rightarrow$ W witness

iii) $\mathrm{tr}(W\rho)<0 \Rightarrow (\Lambda\otimes\mathbb{I})(\rho)\ngeq 0.$

Proof of i): Let $\rho\geq 0$. Then,

$$\langle\phi|\Lambda(\rho)^T|\phi\rangle = d\langle\phi|\,\mathrm{tr}_B\left(W(\mathbb{I}\otimes\rho)\right)|\phi\rangle$$

$$= d\,\mathrm{tr}\left[W\underbrace{\left(|\phi\rangle\langle\phi|\otimes\rho\right)}_{\text{sep. state!}}\right]\geq 0$$

(W witness!)

$\Rightarrow$ Λ is a positive map!

For ii) and iii), use

$$\mathrm{tr}\left[W(A\otimes B)\right] = \mathrm{tr}_A\left[\mathrm{tr}_B\left(W(\mathbb{I}\otimes B)\right)\cdot A\right]$$

$$= \frac{1}{d}\,\mathrm{tr}_A\left[\Lambda(B)^T A\right] = \frac{1}{d}\sum_{ij}\left[\Lambda(B)^T\right]_{ij} A_{ji}$$

$$= \langle\Omega|\,A\otimes\Lambda(B)\,|\Omega\rangle$$

Proof of ii) Let $\rho_i^A, \rho_i^B \geq 0$; $\rho_i \geq 0$. Then,

$$\mathrm{tr}\left[W\left(\sum \rho_i \rho_i^A \otimes \rho_i^B\right)\right] = \sum \rho_i \, \mathrm{tr}\left[W\left(\rho_i^A \otimes \rho_i^B\right)\right]$$

$$\overset{\circledast}{=} \sum \rho_i \langle \Omega | \underbrace{\rho_i^A}_{\geq 0} \otimes \underbrace{\Lambda(\rho_i^B)}_{\geq 0, \text{ as } \Lambda \text{ positive.}} | \Omega \rangle \geq 0$$

$\Rightarrow W$ is witness.

Proof of iii): $\circledast$ together with linearity $\Rightarrow$

$$\Rightarrow \mathrm{tr}(W\rho) = \langle \Omega | (I \otimes \Lambda)(\rho) | \Omega \rangle$$

which implies

$$\mathrm{tr}(W\rho) < 0 \Rightarrow (I \otimes \Lambda)(\rho) \neq 0$$

$\Rightarrow \Lambda$ detects all states which W detects. ∎

Conversely: Checking the witness W amounts to apply the positive map Λ, but checking for negative eigenvalues only along $|\Omega\rangle$ $\Rightarrow$ weaker criteria!

Corollary: A state is separable if and only if

$(\Lambda \otimes I)(\rho) \geq 0 \quad \forall$ positive maps Λ

(since sep. states $\longleftrightarrow$ witnesses $\longleftrightarrow$ positive maps).

Example: $W = \mathbb{F}$.

$$\Lambda(X) = d\, \mathrm{tr}_B\left(\mathbb{F}(I \otimes X^T)\right) = d \cdot X^T \cdot I = d X^T$$

↑ "magic formula" for partial trace!

$\Longrightarrow W = \mathbb{F}$ corresponds to PPT criterion!

(but: PPT detect all Bell states, $\mathbb{F}$ only the antisym. state)

e) Quantification of mixed state entanglement

How can we quantify entanglement of mixed states?

i) Entanglement needed to create the state, e.g. minimal amount of $E(|\psi\rangle) = S(\mathrm{tr}_B |\psi\rangle\langle\psi|)$ needed:

$$E_F(\rho) := \min_{\{p_i, |\psi_i\rangle\}} \sum p_i E(|\psi_i\rangle)$$
$$\text{s.t. } \sum p_i |\psi_i\rangle\langle\psi_i| = \rho$$

"entanglement of formation"

or an asymptotic version (cost per copy)

$$E_C(\rho) := \lim_{N\to\infty} \frac{1}{N} E_F(\rho^{\otimes N})$$

"entanglement cost"

Already E_F very hard to compute – need to minimize concave function $S(\rho)$ over convex set

of decompositions $\rho = \sum p_i \rho_i$ – i.e. minimum obtained at boundary (hard!)

Also, $E_F(\rho \otimes \sigma) \neq E_F(\rho) + E_F(\sigma)$

(additivity of E_F – counterex. exist (Hastings))

But: Analytic formula for $d_A \times d_B = 2 \times 2$ for E_F exists (Wootters), based on so-called "concurrence".

ii) Extractable entanglement:

"Distillable entanglement" $E_D(\rho)$

$E_D(\rho)$ = max. asympt. rate $R = \frac{m}{N}$ achievable for LOCC-protocol $\mathcal{E}_N$: $\| \mathcal{E}_N(\rho^{\otimes N}) - |\phi^+\rangle\langle\phi^+|^{\otimes m} \| \to 0$

↑ suitable dist. meas, typ. trace norm.

as $N, m \to \infty$.

Even harder to compute:

Asymptotic ($N, m \to \infty$) and any # of LOCC rounds.

(Versions w/ restricted LOCC rounds exist, e.g. me-way dist. entanglement, ...)

Observation: $E_F(\rho) \geq E_C(\rho) \geq E_D(\rho)$

Generally, $E_C(\rho) \neq E_D(\rho)$: for most states, process is not reversible.

$\Rightarrow$ no unique measure as for pure states!

Example: ρ PPT, i.e. $\rho^{T_A} \geq 0$.

LOCC preserves PPT $\Rightarrow$ PPT states are undistillable, $E_D(\rho) = 0$.

But there exist PPT states with $E_C(\rho) > 0$:

"PPT bound entangled states"

(Note: The converse problem -

does $\rho^{T_A} \neq 0 \Rightarrow E_D(\rho) > 0$ hold?

- is a big open problem - the existence of NPT bound entangled states.)

Problem: These might be natural ent. measures, but they are essentially impossible to compute.

→ **Computable ent. measure** desirable!

Wishlist for a good ent. measure:

- **LOCC-monotone:** Cannot be increased by LOCC (probably most relevant!)
- $E(\rho) = 0$ for sep. states ρ (and only these?)
- **additive:** $E(\rho \otimes \sigma) = E(\rho) + E(\sigma)$
- **continuous:** $\rho \approx \sigma \Rightarrow E(\rho) \approx E(\sigma)$
  (typ. trace norm)
- $E_D \leq E \leq E_C$.
- coincides with $E(|\psi\rangle) = S(\mathrm{tr}_B |\psi\rangle\langle\psi|)$ on pure states.
- computable?

(Almost) impossible to get all – LOCC monotonicity is probably the most relevant one.

Negativity – a computable entanglement measure

Found previously:

ρ^{T_A} has neg. eigenvalues $\Rightarrow$ ρ entangled.

Use negative eigenvalues to measure entanglement:

"Negativity" $\mathcal{N}(\rho) = \frac{1}{2}\left(\sum_i |\lambda_i(\rho^{T_A})| - 1\right)$

(λ_i: eigenvalues)

$\sum_i |\lambda_i(\rho^{T_A})| =: \|\rho^{T_A}\|_1$: trace norm

$$= \frac{1}{2}\left(\|\rho^{T_A}\|_1 - 1\right)$$

$$= -\sum_{\text{neg. eigenvalues}} \lambda_i(\rho^{T_A})$$

holds since $\sum \lambda_i(\rho^{T_A}) = \mathrm{tr}(\rho^{T_A}) = \mathrm{tr}(\rho) = 1$

or "log-negativity"

$$E_N(\rho) = \log_2 \|\rho^{T_A}\|_1$$

Properties:

Negativity $\mathcal{N}$:

- LOCC – monotone
- not additive
- $\mathcal{N}(\rho) = 0$ for ρ separable,
  but also $\mathcal{N}(\rho)$ on PPT ent. states!
- $\neq E(|\psi\rangle)$ for pure states
- Continuous

Log-negativity E_N:

- not an LOCC monotone (!)
- additive
- $E_N(\rho) = 0$ for ρ sep, but also on PPT ent. st.
- $\neq E(|\psi\rangle)$ for pure states
- continuous